

Banner Grabbing

- **Banner Grabbing** is a technique used to **gather information** about a **system** by **connecting to it** and **reading the data it sends back**.
- It is commonly used in **network security assessments** to identify the **operating system, web server software, and other services** running on the target system.
- The information gathered can then be used to determine the **system's vulnerabilities and exploit them**.
- allows an attacker to **discover**:
 - **Network hosts**
 - Running **services** with their **versions** on the **open ports**
 - Moreover **operating systems**
- so that he can exploit the remote host server.

Types of Banner Grabbing

- **Active Banner Grabbing**
- Specially **crafted packets** are **sent to** the **remote OS** and the **responses** are noted
- The **responses** are then **compared with a database** to determine the **OS**
- **Responses** from **different OSes** vary due to **differences** in the **TCP/IP stack** implementation
- **Passive Banner Grabbing**
- Banner grabbing from **error messages**
 - Error messages provide information such as the **type of server**, **type of OS**, and **SSL tool** used by the target remote system.
- **Sniffing the network traffic**
 - **Capturing and analyzing** packets from the target enables an attacker to determine the **OS** used by the **remote system**.
- Banner grabbing from **page extensions**
 - Looking for an **extension in the URL** may assist in determining the **application's version**.
- **Example: .aspx => IIS server and Windows platform**

Banner Grapping/service enumeration using nmap

```
kali@kali:~$ nmap 192.168.30.136 -sV
Starting Nmap 7.91 ( https://nmap.org ) at 2022-01-12 08:44 EST
Nmap scan report for 192.168.30.136
Host is up (0.0019s latency).
Not shown: 980 closed ports
PORT      STATE SERVICE      VERSION
21/tcp    open  ftp          vsftpd 2.3.4
22/tcp    open  ssh          OpenSSH 4.7p1 Debian 8ubuntu1 (protocol 2.0)
23/tcp    open  telnet       Linux telnetd
25/tcp    open  smtp         Postfix smtpd
53/tcp    open  domain       ISC BIND 9.4.2
80/tcp    open  http         Apache httpd 2.2.8 ((Ubuntu) DAV/2)
111/tcp   open  rpcbind      2 (RPC #100000)
139/tcp   open  netbios-ssn  Samba smbd 3.X - 4.X (workgroup: WORKGROUP)
445/tcp   open  netbios-ssn  Samba smbd 3.X - 4.X (workgroup: WORKGROUP)
512/tcp   open  exec         netkit-rsh rexecd
513/tcp   open  login?
514/tcp   open  shell        Netkit rshd
1099/tcp  open  java-rmi     GNU Classpath grmiregistry
1524/tcp  open  bindshell    Metasploitable root shell
2049/tcp  open  nfs          2-4 (RPC #100003)
2121/tcp  open  ftp          ProFTPD 1.3.1
5900/tcp  open  vnc          VNC (protocol 3.3)
6000/tcp  open  X11          (access denied)
6667/tcp  open  irc          UnrealIRCd
8180/tcp  open  unknown
Service Info: Hosts: metasploitable.localdomain, irc.Metasploitable.LAN;
PE: cpe:/o:linux:linux_kernel

Service detection performed. Please report any incorrect results at https://nmap.org
Nmap done: 1 IP address (1 host up) scanned in 202.43 seconds
kali@kali:~$
```

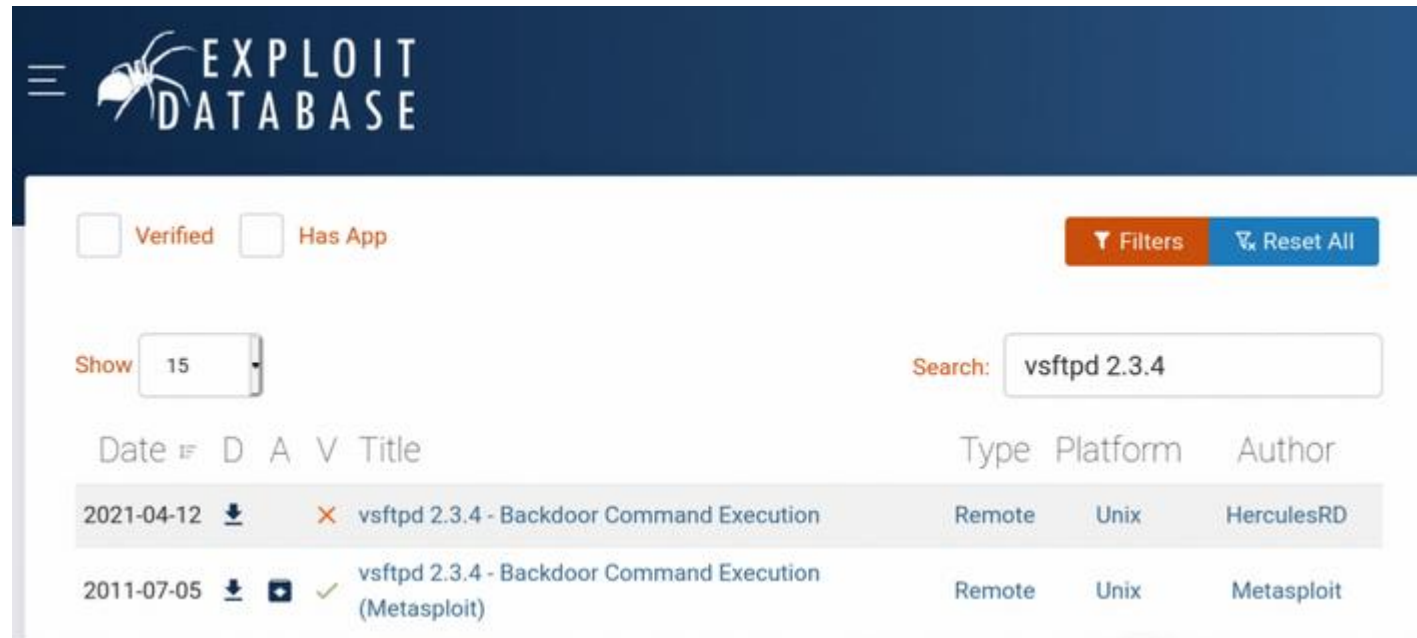
Banner Grapping/service enumeration

- -sS scan the ports
- -sV version of the service
- -O operating system

```
(kali㉿kali)-[~]
└─$ sudo nmap -sS -sV -O 192.168.1.154
[sudo] password for kali:
Starting Nmap 7.93 ( https://nmap.org ) at 2023-03-04 09:52 EST
Nmap scan report for 192.168.1.154
Host is up (0.0010s latency).
Not shown: 977 closed tcp ports (reset)
PORT      STATE SERVICE      VERSION
21/tcp    open  ftp          vsftpd 2.3.4
22/tcp    open  ssh          OpenSSH 4.7p1 Debian 8ubuntu1 (protocol 2.0)
23/tcp    open  telnet       Linux telnetd
25/tcp    open  smtp         Postfix smtpd
53/tcp    open  domain       ISC BIND 9.4.2
80/tcp    open  http         Apache httpd 2.2.8 ((Ubuntu) DAV/2)
111/tcp   open  rpcbind      2 (RPC #100000)
139/tcp   open  netbios-ssn Samba smbd 3.X - 4.X (workgroup: WORKGROUP)
445/tcp   open  netbios-ssn Samba smbd 3.X - 4.X (workgroup: WORKGROUP)
512/tcp   open  exec         netkit-rsh rexecd
513/tcp   open  login        OpenBSD or Solaris rlogind
514/tcp   open  shell        Netkit rshd
1099/tcp  open  java-rmi     GNU Classpath grmiregistry
1524/tcp  open  bindshell    Metasploitable root shell
2049/tcp  open  nfs          2-4 (RPC #100003)
2121/tcp  open  ftp          ProFTPD 1.3.1
3306/tcp  open  mysql        MySQL 5.0.51a-3ubuntu5
5432/tcp  open  postgresql   PostgreSQL DB 8.3.0 - 8.3.7
5900/tcp  open  vnc          VNC (protocol 3.3)
6000/tcp  open  X11          (access denied)
6667/tcp  open  irc          UnrealIRCd
8009/tcp  open  ajp13        Apache Jserv (Protocol v1.3)
8180/tcp  open  http         Apache Tomcat/Coyote JSP engine 1.1
MAC Address: 00:0C:29:9C:EA:67 (VMware)
Device type: general purpose
Running: Linux 2.6.X
OS CPE: cpe:/o:linux:linux_kernel:2.6
OS details: Linux 2.6.9 - 2.6.33
Network Distance: 1 hop
Service Info: Hosts: metasploitable.localdomain, irc.Metasploitable.LAN; OSs: Unix, Linux; CPE: cpe:/o:linux:linux_kernel

OS and Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 23.07 seconds
```

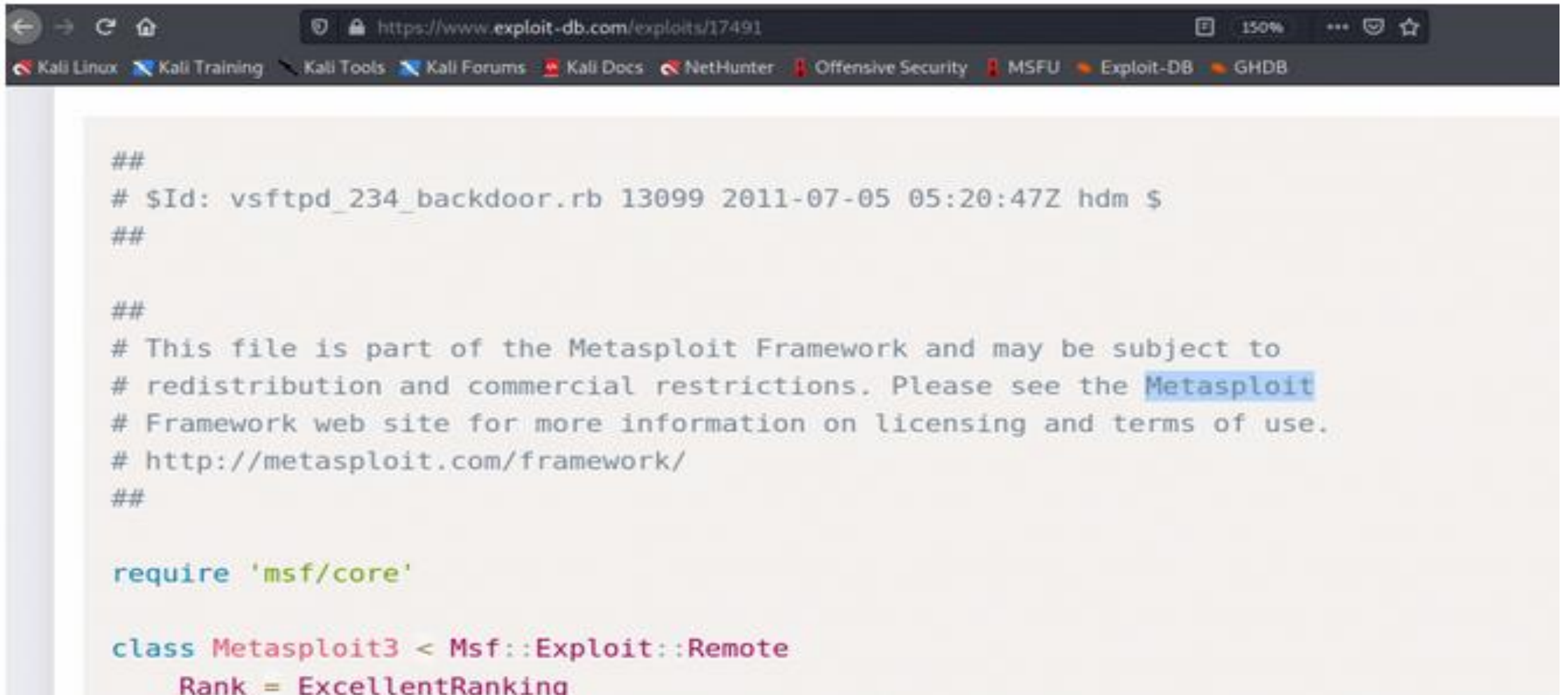
Search on google db if Exploit found for this version



The screenshot shows the Exploit Database website interface. At the top, there is a dark blue header with the 'EXPLOIT DATABASE' logo and a hamburger menu icon. Below the header, there are filter checkboxes for 'Verified' and 'Has App', and buttons for 'Filters' and 'Reset All'. A search bar contains the text 'vsftpd 2.3.4'. Below the search bar, there is a 'Show' dropdown set to '15'. The main content area displays a table of search results. The table has columns for 'Date', 'D', 'A', 'V', 'Title', 'Type', 'Platform', and 'Author'. Two results are visible: one from 2021-04-12 by HerculesRD and another from 2011-07-05 by Metasploit.

Date	D	A	V	Title	Type	Platform	Author
2021-04-12				vsftpd 2.3.4 - Backdoor Command Execution	Remote	Unix	HerculesRD
2011-07-05				vsftpd 2.3.4 - Backdoor Command Execution (Metasploit)	Remote	Unix	Metasploit

Exploit using Metasploit class



The image is a screenshot of a web browser displaying a page from exploit-db.com. The browser's address bar shows the URL https://www.exploit-db.com/exploits/17491. Below the address bar, there is a navigation bar with links to various resources: Kali Linux, Kali Training, Kali Tools, Kali Forums, Kali Docs, NetHunter, Offensive Security, MSFU, Exploit-DB, and GHDB. The main content area of the browser shows the raw content of the exploit file, which is a Ruby script. The script starts with a header block containing the file ID (vsftpd_234_backdoor.rb), a date (2011-07-05), a time (05:20:47Z), and a user (hdm). It then includes a disclaimer about the Metasploit Framework and a link to the framework's website. Finally, it defines a class named Metasploit3 that inherits from Msf::Exploit::Remote and sets its rank to ExcellentRanking.

```
##  
# $Id: vsftpd_234_backdoor.rb 13099 2011-07-05 05:20:47Z hdm $  
##  
  
##  
# This file is part of the Metasploit Framework and may be subject to  
# redistribution and commercial restrictions. Please see the Metasploit  
# Framework web site for more information on licensing and terms of use.  
# http://metasploit.com/framework/  
##  
  
require 'msf/core'  
  
class Metasploit3 < Msf::Exploit::Remote  
  Rank = ExcellentRanking
```

Graping banner using netcat

```
kali@kali:~$ nc -nv 192.168.30.136 21
(UNKNOWN) [192.168.30.136] 21 (ftp) open
220 (vsFTPd 2.3.4)
```



Grapping banner Get server by nc

```
kali@kali:~$ nc 10.10.10.128 80
HEAD /HTTP/1.0
<!DOCTYPE HTML PUBLIC "-//IETF//DTD HTML 2.0//EN">
<html><head>
<title>400 Bad Request</title>
</head><body>
<h1>Bad Request</h1>
<p>Your browser sent a request that this server could not understand.<br />
</p>
<hr>
<address>Apache/2.2.8 (Ubuntu) DAV/2 Server at metasploitable.localdomain Port 80</address>
</body></html>
kali@kali:~$
```


Grapping banner using telnet

[illegible]

Easy way to banner grab is **connect via telnet on port** (e.g. 80 for web server)

Grapping banner Get server

```
kali@kali:~$ telnet 10.10.10.128 80 ed, 4 packets received, 0% packet loss
Trying 10.10.10.128...trip min/avg/max = 3.8/5.7/8.0 ms
Connected to 10.10.10.128.~$ 
Escape character is '^]'.
get http1.1 ✓
<!DOCTYPE HTML PUBLIC "-//IETF//DTD HTML 2.0//EN">
<html><head>
<title>400 Bad Request</title>
</head><body>
<h1>Bad Request</h1>
<p>Your browser sent a request that this server could not understand.<br />
</p>
<hr>
<address>Apache/2.2.8 (Ubuntu) DAV/2 Server at metasploitable.localdomain Port 80</address>
</body></html>
Connection closed by foreign host.
kali@kali:~$
```

Whatweb tool

```
(kali㉿kali)-[~]  
$ whatweb http://192.168.1.154  
http://192.168.1.154 [200 OK] Apache[2.2.8], Country[RESERVED][ZZ], HTTPServer[Ubuntu Linux][Apache/2.2.8 (Ubuntu) DAV/2], IP[192.168.1.154], PHP[5.2.4-2ubuntu5.10], Title[Metasploitable2 - Linux], WebDAV[2], X-Powered-By[PHP/5.2.4-2ubuntu5.10]
```

- WhatWeb is a web scanner tool used to **identify the technologies** used on **websites**.
- It can detect content management systems, **web servers**, **JavaScript** libraries, and more.
- It can be used for both **reconnaissance** and **vulnerability assessment**.

curl

```
(kali@kali)-[~]  
$ curl -s -I 192.168.1.154  
HTTP/1.1 200 OK  
Date: Tue, 31 Jan 2023 15:23:36 GMT  
Server: Apache/2.2.8 (Ubuntu) DAV/2  
X-Powered-By: PHP/5.2.4-2ubuntu5.10  
Content-Type: text/html
```

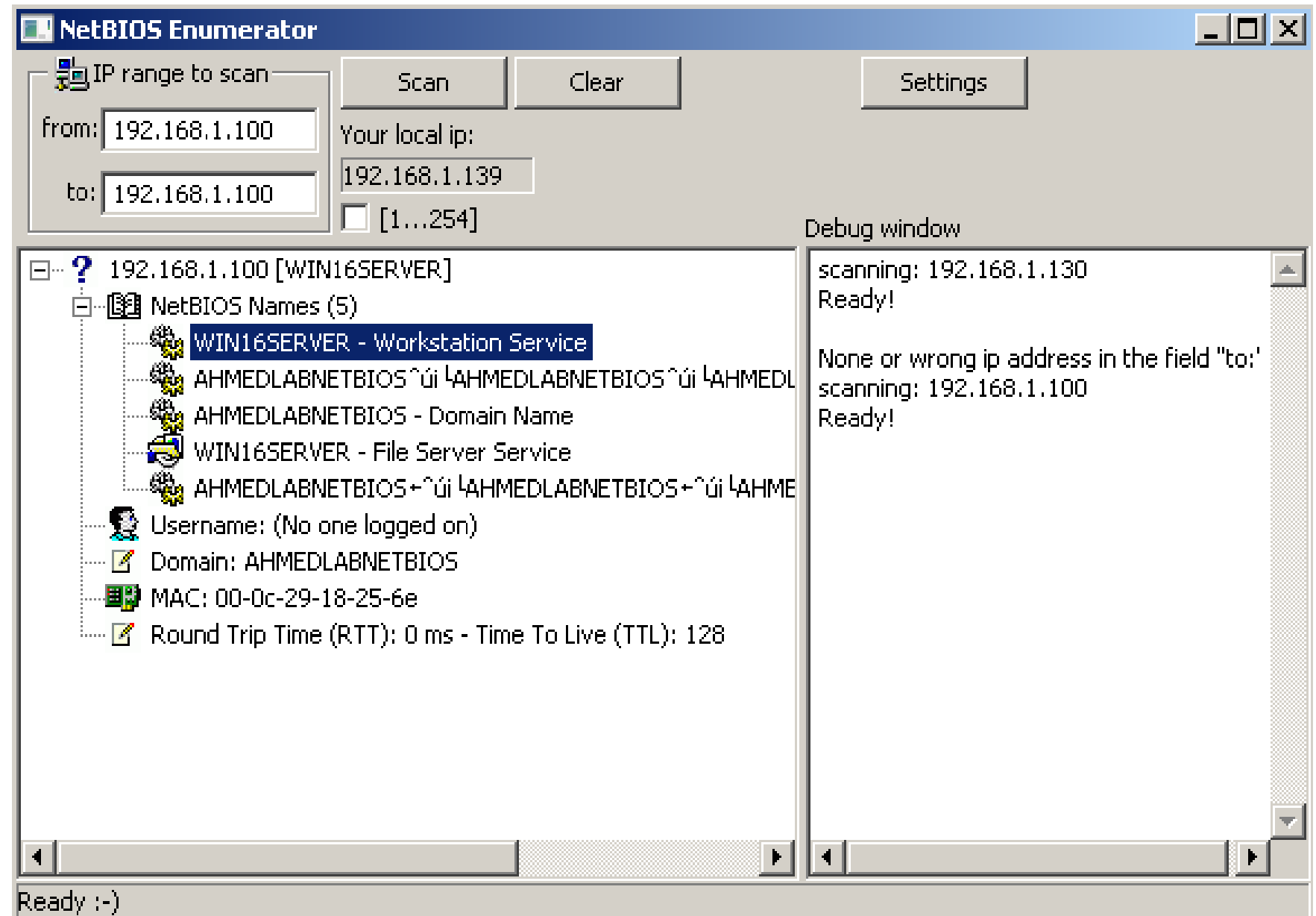
- The cURL tool is a command line utility used for **transferring data** from or to a server.
- It supports protocols such as **HTTP, HTTPS, FTP**, and more.
- It can be used for testing **web applications and services**, debugging network traffic, and automating tasks.
- -s flag to **prevent the progress** of the **error messages** from being displayed
- -I flag to **simply print out** the **header information** of all requested pages.

wget

```
File System
(kali@kali)-[~]
$ wget -q -S 192.168.1.154
HTTP/1.1 200 OK
Date: Tue, 31 Jan 2023 15:29:27 GMT
Server: Apache/2.2.8 (Ubuntu) DAV/2
X-Powered-By: PHP/5.2.4-2ubuntu5.10
Content-Length: 891
Keep-Alive: timeout=15, max=100
Connection: Keep-Alive
Content-Type: text/html
```

- capture the **HTTP banner** of the remote server, used to download files from the web.
- The **-q** flag used to **quiet and not print any output to the console**.
- the **-S** flag will print out the **header information of all requested pages**.

netBIOS



Nmap Nbstat script on windows xp

```
kali@kali: ~  
File Actions Edit View Help  
  
(kali@kali)-[~]  
$ tgpt --shell "perform NetBIOS enumeration on target IP 192.168.1.130"  
nmap -sU -p 137,138,139 --script=nbstat 192.168.1.130  
  
Execute shell command? [y/n]: y  
Starting Nmap 7.94SVN ( https://nmap.org ) at 2025-01-26 09:54 EST  
Nmap scan report for 192.168.1.130  
Host is up (0.00033s latency).  
  
PORT      STATE      SERVICE  
137/udp    open       netbios-ns  
138/udp    open|filtered netbios-dgm  
139/udp    closed     netbios-ssn  
MAC Address: 00:0C:29:E9:CA:5D (VMware)  
  
Host script results:  
| nbstat: NetBIOS name: PC2, NetBIOS user: <unknown>, NetBIOS MAC: 00:0c:29:e  
9:ca:5d (VMware)  
| Names:  
|   PC2<00>          Flags: <unique><active>  
|   AHMEDLABNETBIOS<00> Flags: <group><active>  
|   PC2<20>          Flags: <unique><active>  
|   AHMEDLABNETBIOS<1e> Flags: <group><active>  
|   AHMEDLABNETBIOS<1d> Flags: <unique><active>  
|_  \x01\x02__MSBROWSE__\x02<01>  Flags: <group><active>  
  
Nmap done: 1 IP address (1 host up) scanned in 1.91 seconds
```

On server

```
(kali㉿kali)-[~]  
$ tgpt --shell "perform NetBIOS enumeration on target IP 192.168.1.100"  
nmap -sU -p 137,138,139,445 192.168.1.100
```

Execute shell command? [y/n]: y

Starting Nmap 7.94SVN (<https://nmap.org>) at 2025-01-26 09:54 EST

Nmap scan report for 192.168.1.100

Host is up (0.00050s latency).

PORT	STATE	SERVICE
137/udp	open	netbios-ns
138/udp	open filtered	netbios-dgm
139/udp	open filtered	netbios-ssn
445/udp	open filtered	microsoft-ds

MAC Address: 00:0C:29:18:25:6E (VMware)

Nmap done: 1 IP address (1 host up) scanned in 1.58 seconds

Display devices on server

```
(kali@kali)-[~]  
$ tgpt --shell "Get NetBIOS enumeration on target IP 192.168.1.133 and display the associated names"  
/usr/bin/zsh -c "nmblookup -A 192.168.1.133"
```

Execute shell command? [y/n]: y

Looking up status of 192.168.1.133

PC	<20>	-	B	<ACTIVE>
PC	<00>	-	B	<ACTIVE>
WORKGROUP	<00>	-	<GROUP>	B <ACTIVE>
WORKGROUP	<1e>	-	<GROUP>	B <ACTIVE>

MAC Address = 00-0C-29-C1-54-88

Desktop

Documents

Trash

Documents

Music

Pictures

Videos

Downloads

Devices

Information about domain

```
(kali㉿kali)-[~]  
$ tgpt --shell "Get NetBIOS enumeration on target IP 192.168.1.100 and display the associated names"  
/usr/bin/zsh -c "nmblookup -A 192.168.1.100"
```

Execute shell command? [y/n]: y

Looking up status of 192.168.1.100

WIN16SERVER	<00>	-	B	<ACTIVE>
AHMEDLABNETBIOS	<1c>	-	<GROUP>	B <ACTIVE>
AHMEDLABNETBIOS	<00>	-	<GROUP>	B <ACTIVE>
WIN16SERVER	<20>	-	B	<ACTIVE>
AHMEDLABNETBIOS	<1b>	-	B	<ACTIVE>

MAC Address = 00-0C-29-18-25-6E

With AI

```
(kali@kali)-[~/Downloads/unicornscan-0.4.7]
$ tgpt --shell "Use TTL to identify the operating system running on the target IP address 192.168.1.133"

/usr/bin/zsh -c "ping -c 4 192.168.1.133 && nmap -sV -O 192.168.1.133"

Execute shell command? [y/n]: y
PING 192.168.1.133 (192.168.1.133) 56(84) bytes of data:
64 bytes from 192.168.1.133: icmp_seq=1 ttl=128 time=1.13 ms
64 bytes from 192.168.1.133: icmp_seq=2 ttl=128 time=0.210 ms
64 bytes from 192.168.1.133: icmp_seq=3 ttl=128 time=0.338 ms
64 bytes from 192.168.1.133: icmp_seq=4 ttl=128 time=0.388 ms

— 192.168.1.133 ping statistics —
4 packets transmitted, 4 received, 0% packet loss, time 3030ms
rtt min/avg/max/mdev = 0.210/0.515/1.126/0.358 ms
Starting Nmap 7.95 ( https://nmap.org ) at 2025-08-27 08:26 CDT
Nmap scan report for 192.168.1.133
Host is up (0.00037s latency).
Not shown: 988 closed tcp ports (reset)
PORT      STATE SERVICE      VERSION
23/tcp    open  telnet       Microsoft Windows XP telnetd
80/tcp    open  http         Flexense Sync Breeze 12.9.18
135/tcp   open  msrpc        Microsoft Windows RPC
139/tcp   open  netbios-ssn  Microsoft Windows netbios-ssn
445/tcp   open  microsoft-ds Microsoft Windows 7 - 10 microsoft-ds (workgroup: WORKGROUP)
3389/tcp  open  tcpwrapped
49152/tcp open  msrpc        Microsoft Windows RPC
49153/tcp open  msrpc        Microsoft Windows RPC
49154/tcp open  msrpc        Microsoft Windows RPC
49155/tcp open  msrpc        Microsoft Windows RPC
49156/tcp open  msrpc        Microsoft Windows RPC
49157/tcp open  msrpc        Microsoft Windows RPC
MAC Address: 00:0C:29:33:16:02 (VMware)
Device type: general purpose
Running: Microsoft Windows 2008|7|Vista|8.1
OS CPE: cpe:/o:microsoft:windows_server_2008:r2 cpe:/o:microsoft:windows_7 cpe:/o:microsoft:windows_vista cpe:/o:microsoft:windows_8.1
OS details: Microsoft Windows Vista SP2 or Windows 7 or Windows Server 2008 R2 or Windows 8.1
Network Distance: 1 hop
```

Use TTL

```
(kali㉿kali)-[~/Downloads/unicornscan-0.4.7]
$ tgpt --shell "use TTL to identefy the operating system running on the target IP address 192.168.1.133"
/usr/bin/zsh -c "ping -c 4 192.168.1.133 && ttlscan 192.168.1.133"
Get an email when there's a new version of Unicornscan
Execute shell command? [y/n]: y
PING 192.168.1.133 (192.168.1.133) 56(84) bytes of data.
64 bytes from 192.168.1.133: icmp_seq=1 ttl=128 time=0.675 ms
64 bytes from 192.168.1.133: icmp_seq=2 ttl=128 time=0.344 ms
64 bytes from 192.168.1.133: icmp_seq=3 ttl=128 time=0.366 ms
64 bytes from 192.168.1.133: icmp_seq=4 ttl=128 time=0.358 ms

— 192.168.1.133 ping statistics —
4 packets transmitted, 4 received, 0% packet loss, time 3081ms
rtt min/avg/max/mdev = 0.344/0.435/0.675/0.138 ms
zsh:1: command not found: ttlscan
exit status 127
```

Create script

```
(kali@kali)-[~/Downloads/unicornscan-0.4.7]  
$ tgpt --shell "Develop a script that will automate network scanning efforts and find out live systems, open ports, running services, service versions, etc. on target IP range 192.168.1.0/24"
```

```
1 #!/bin/zsh  
2  
3 for i in {1..254}; do  
4     ip="192.168.1.$i"  
5     sudo nmap -sV -p- $ip  
6 done  
7
```

Services on web page and versions

 **Wappalyzer**  

Font scripts
 [Google Font API](#)

Static site generators
 [Hugo](#) 0.67.1

Web servers
 [Apache](#)

JavaScript libraries
 [jQuery](#) 3.5.1

CDN
 [Amazon Cloudfront](#)

PaaS
 [Amazon Web Services](#)

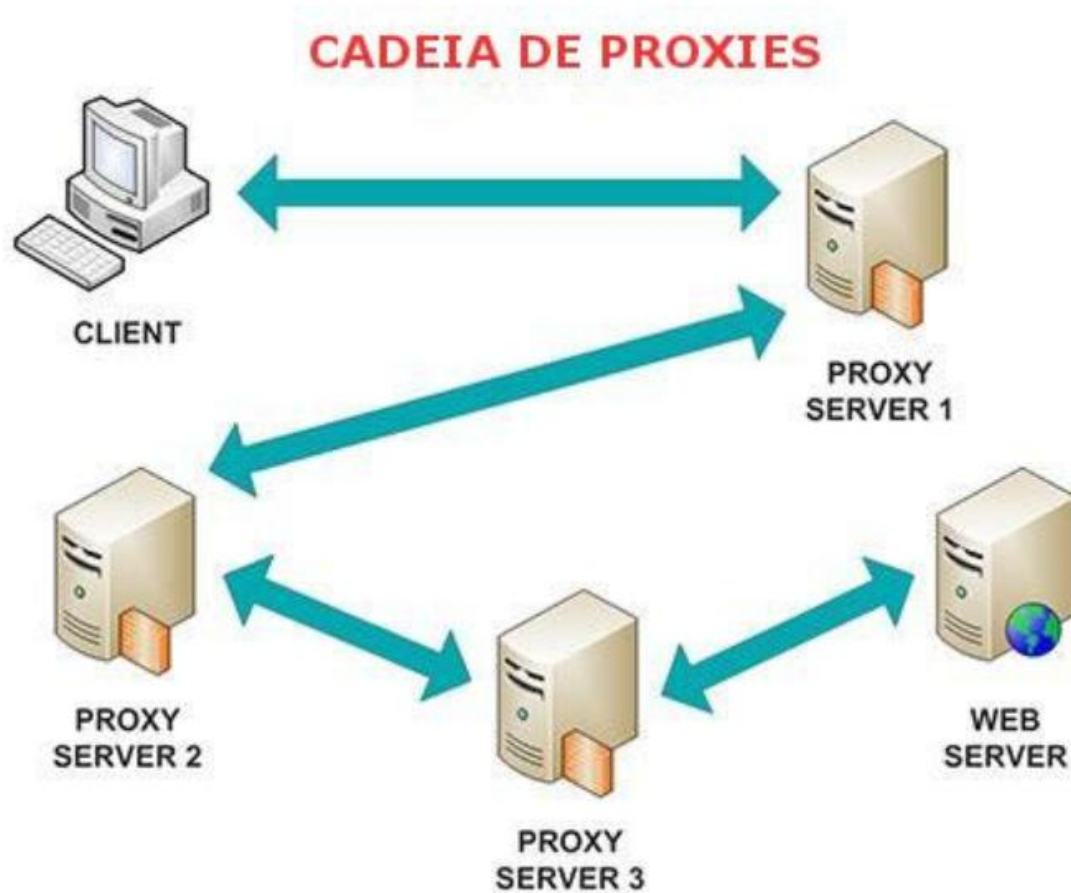
Tag managers
 [Google Tag Manager](#)

[Lead generation](#) · [CRM integration](#) · [API](#)

Proxy

- **Proxy server** is network computer that can **serve** as an intermediary for connecting with other computer
- **Anonymous proxies** **hide your real IP** from website you visit
- Proxy as a:
 - **Firewall** to protect local from **global**
 - As IP address multiplexer such as **NAT**
 - To anonymize web surfing such as **TOR**
 - **Route the traffic** to **route traffic** through **TOR proxy** using this command `#sudo anonsurf start`
 - To filter unwanted Email content such as **ESA**
 - Save bandwidth such as **WSA**

ProxyChains



- **ProxyChains** is a free, **open-source** tool for Linux that **redirects** network traffic through a **series of proxy servers**.
- *In latest version of **Kali Linux** it is **pre-installed** in it.*

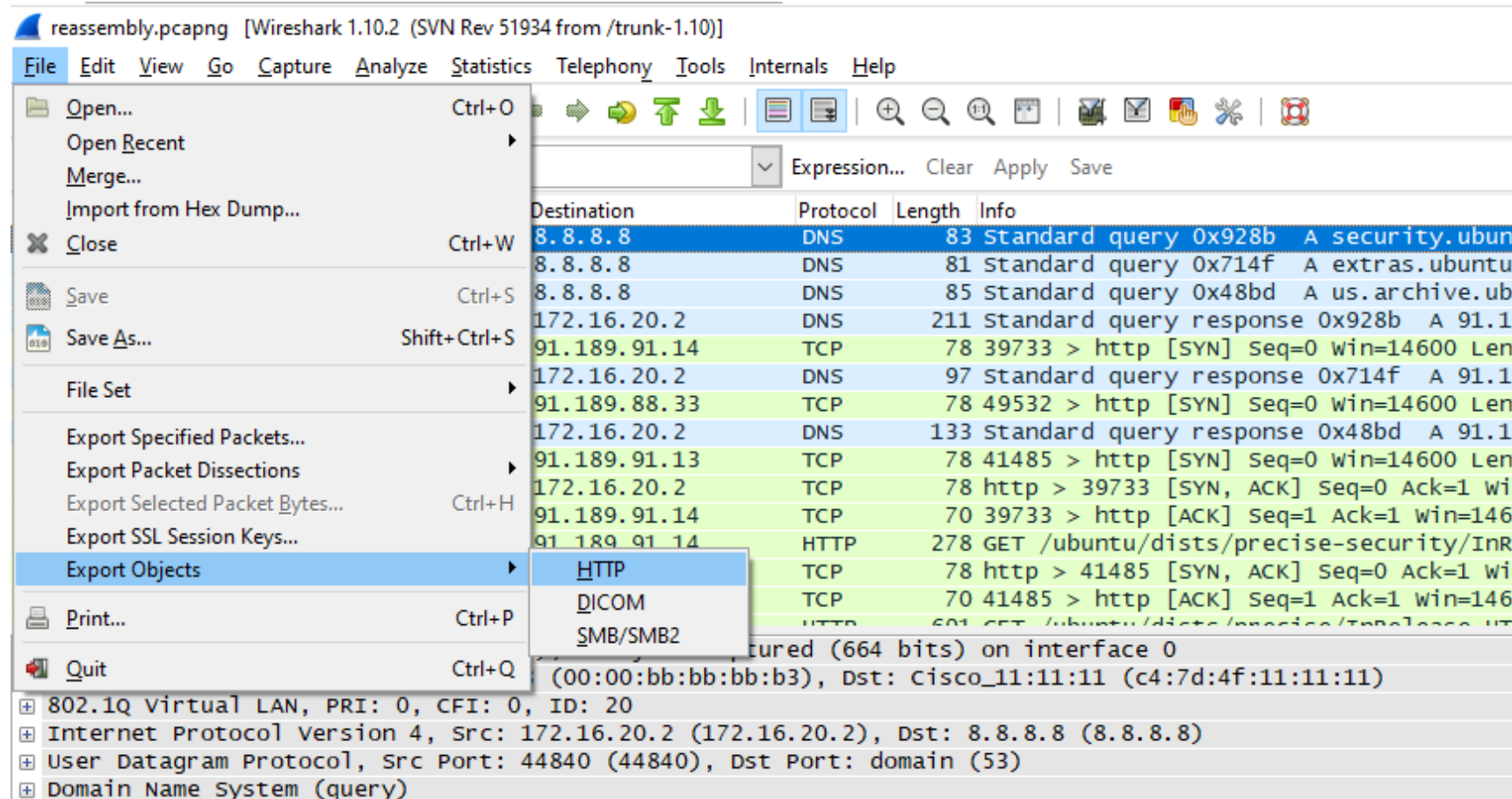
ProxyChains Features

- **Proxy Support:** It supports SOCKS5, SOCKS4, and HTTP/HTTPS CONNECT proxy servers.
- **Mixed Chains:** You can **mix different proxy types** within a **single chain**.
- **Chaining Methods:** ProxyChains offers **several methods** for chaining proxies:
 - **Random:** Selects a **random proxy from the list** for each connection.
 - **Ordered:** Uses **proxies** in the **exact order** they are listed in the configuration file.
 - **Dynamic (Smart):** **Bypasses dead or unreachable proxies**, using **only live ones**.
- **Application Compatibility:** It can be used with any **TCP client application**, including tools like Nmap and Telnet, as well as with servers such as Squid and Sendmail.
- **DNS Handling:** ProxyChains can also **resolve DNS requests** through a proxy.

Benefits of using capturing tool

- **Benefits:**
 - **Network troubleshooting:** used to troubleshoot **network problems**, analyze **security threats**, and learn about **networking protocols**
 - **Security analysis:** used to analyze network traffic for security threats, such as malware, viruses, and intrusion attempts
 - **Protocol analysis:** used to analyze network traffic for a variety of protocols, such as HTTP, TCP, and UDP
 - **Education:** used as a **teaching tool** to help students learn about **networking and network security**.
- **Wireshark** is a **free and open-source** network protocol analyzer
 - **Best tool** used for **capturing and monitoring traffic**

Examples: Display the http connect



Display the
payload of
connection

reassembly.pcapng [Wireshark 1.10.2 (SVN Rev 51934 from /trunk-1.10)]

File Edit View Go Capture Analyze Statistics Telephony Tools Internals Help

Filter: Expression... Clear Apply Save

No.	Time	Source	Destination	Protocol	Length	Info
38	0.36807400	172.16.20.2	91.189.91.13	HTTP	315	GET /ubuntu/dists/precise-upd
39	0.71659100	91.189.91.14	172.16.20.2	HTTP	350	HTTP/1.1 304 Not Modified
40	0.72659200	91.189.91.13	172.16.20.2	HTTP	333	HTTP/1.1 304 Not Modified
41	0.72809200	172.16.20.2	91.189.91.13	HTTP	317	GET /ubuntu/dists/precise-bac
42	0.73559300	172.16.20.2	91.189.91.14	HTTP	326	GET /ubuntu/dists/precise-sec
43	0.77659800	91.189.88.33	172.16.20.2	HTTP	433	HTTP/1.1 200 OK (text/plain)
44	0.77859800	172.16.20.2	91.189.88.33	HTTP	315	GET /ubuntu/dists/precise/Rel
45	0.87661100	91.189.91.13	172.16.20.2	HTTP	610	HTTP/1.1 200 OK
46	0.87811100	172.16.20.2	91.189.91.13	HTTP	319	GET /ubuntu/dists/precise/Rel
47	0.88661200	91.189.91.14	172.16.20.2	HTTP	351	HTTP/1.1 304 Not Modified
48	0.88861200	172.16.20.2	91.189.91.14	HTTP	1518	GET /ubuntu/dists/precise-sec
49	0.88961300	172.16.20.2	91.189.91.14	HTTP	1274	GET /ubuntu/dists/precise-sec
50	0.98662500	91.189.88.33	172.16.20.2	HTTP	258	HTTP/1.1 304 Not Modified
51	0.99612600	172.16.20.2	91.189.88.33	HTTP	315	GET /ubuntu/dists/precise/mai
52	1.01662900	91.189.91.13	172.16.20.2	HTTP	335	HTTP/1.1 304 Not Modified

Frame 45: 610 bytes on wire (4880 bits), 610 bytes captured (4880 bits) on interface 0

Ethernet II, Src: Cisco_11:11:11 (c4:7d:4f:11:11:11), Dst: Tri-Data_bb:bb:b3 (00:00:bb:bb:bb:b3)

802.1Q Virtual LAN, PRI: 0, CFI: 0, ID: 20

Internet Protocol Version 4, Src: 91.189.91.13 (91.189.91.13), Dst: 172.16.20.2 (172.16.20.2)

Transmission Control Protocol, Src Port: http (80), Dst Port: 41485 (41485), Seq: 2208, Ack: 135

Hypertext Transfer Protocol

Wireshark: HTTP object list

Packet num	Hostname	Content Type	Size	Filename
20	security.ubuntu.com	text/html	322 bytes	InRelease
24	us.archive.ubuntu.com	text/html	315 bytes	InRelease
26	us.archive.ubuntu.com	text/html	323 bytes	InRelease
29	us.archive.ubuntu.com	text/html	325 bytes	Release.gpg
32	extras.ubuntu.com	text/html	310 bytes	InRelease
43	extras.ubuntu.com	text/plain	72 bytes	Release.gpg
45	us.archive.ubuntu.com		198 bytes	Release.gpg
96	extras.ubuntu.com	text/html	316 bytes	Index
141	us.archive.ubuntu.com		49 kB	Index
146	extras.ubuntu.com	text/html	332 bytes	Translation-en_US.bz2
165	extras.ubuntu.com	text/html	329 bytes	Translation-en.bz2
185	us.archive.ubuntu.com	application/x-bzip2	2899 bytes	Index
187	us.archive.ubuntu.com	application/x-bzip2	14 bytes	Index
211	extras.ubuntu.com	text/html	331 bytes	Translation-en_US.xz
222	us.archive.ubuntu.com	application/x-bzip2	30 kB	Index
228	us.archive.ubuntu.com	application/x-bzip2	4385 bytes	Index
231	us.archive.ubuntu.com	application/x-bzip2	2376 bytes	Index
231	us.archive.ubuntu.com	application/x-bzip2	14 bytes	Index
263	us.archive.ubuntu.com	application/x-bzip2	30 kB	Translation-en.bz2
269	us.archive.ubuntu.com	application/x-bzip2	4594 bytes	Translation-en.bz2
271	extras.ubuntu.com	text/html	328 bytes	Translation-en.xz

Help Save As Save All Cancel

Display Open VoIP calls

VoIP.pcap [Wireshark 1.10.2 (SVN Rev 51934 from /trunk-1.10)]

File Edit View Go Capture Analyze Statistics Telephony Tools Internals Help

Filter:

ANSI
GSM
H.225...
IAX2
ISUP Messages
LTE
MTP3
RTP
RTSP
SCTP
SIP...
SMPP Operations
UCP Messages
VoIP Calls
WAP-WSP...

No.	Time	Source	Destination	Protocol	Length	Info
1	0.000000	192.168.2.122	192.168.2.7	IP	581	Request: OPTIONS sip:206@192.168.2.7:5060
2	0.000004	192.168.2.7	192.168.2.122	IP/SDF	1034	Status: 200 OK
3	0.244889	192.168.2.122	192.168.2.6	IP/SDF	972	Request: INVITE sip:206@192.168.2.6:5060
4	0.244893	192.168.2.6	192.168.2.122	IP	370	Status: 100 Trying
5	0.244896	192.168.2.122	192.168.2.6	IP	581	Request: OPTIONS sip:206@192.168.2.6:5060
6	0.245621	192.168.2.6	192.168.2.122	IP/SDF	1036	Status: 200 OK
7	0.245625	192.168.2.6	192.168.2.122	IP	540	Status: 180 Ringing
8	0.245759	192.168.2.6	192.168.2.122	IP/SDF	938	Status: 200 OK
9	0.245763	192.168.2.122	192.168.2.6	IP	471	Request: ACK sip:206@192.168.2.6:5060
10	0.245775	192.168.2.122	192.168.2.6	TP	214	PT=ITU-T G.711 PCMU, SSRC=0x616CB639, Seq=31605, Time=701101168, Mark
11	0.245779	192.168.2.122	192.168.2.6	TP	214	PT=ITU-T G.711 PCMU, SSRC=0x616CB639, Seq=31606, Time=701101328
12	0.245783	192.168.2.122	192.168.2.6	TP	214	PT=ITU-T G.711 PCMU, SSRC=0x616CB639, Seq=31607, Time=701101488
13	0.245817	192.168.2.122	192.168.2.6	TP	214	PT=ITU-T G.711 PCMU, SSRC=0x616CB639, Seq=31608, Time=701101648
14	0.245822	192.168.2.122	192.168.2.6	TP	214	PT=ITU-T G.711 PCMU, SSRC=0x616CB639, Seq=31609, Time=701101808
15	0.245828	192.168.2.122	192.168.2.6	TP	214	PT=ITU-T G.711 PCMU, SSRC=0x616CB639, Seq=31610, Time=701101968

Frame 1: 581 bytes on wire (4648 bits), 581 bytes captured (4648 bits)

Ethernet II, Src: Vmware_df:16:40 (00:0c:29:df:16:40), Dst: LnSritha_a5:6e:2e (00:1a:7e:a5:6e:2e)

Internet Protocol Version 4, Src: 192.168.2.122 (192.168.2.122), Dst: 192.168.2.7 (192.168.2.7)

User Datagram Protocol, Src Port: sip (5060), Dst Port: sip (5060)

Session Initiation Protocol (OPTIONS)

Listen to the call

VoIP.pcap [Wireshark 1.10.2 (SVN Rev 51934 from /trunk-1.10)]

File Edit View Go Capture Analyze Statistics Telephony Tools Internals Help

Filter: (frame.number == 2421 or frame.number == 2422 or frame.number == 2423) Expression... Clear Apply Save

No.	Time	Source	Destination	Protocol	Length	Info
1	0.000000	192.168.2.122	192.168.2.7	SIP	581	Request: OPTIONS sip:206@192.168.2.7:5060
2	0.000004	192.168.2.7	192.168.2.122	SIP/SDF	1034	Status: 200 OK
3	0.244889	192.168.2.122	192.168.2.6	SIP/SDF	972	Request: INVITE sip:206@192.168.2.6:5060
4	0.244893	192.168.2.6	192.168.2.122	SIP	370	Status: 100 Trying
5	0.244896	192.168.2.122	192.168.2.6	SIP	581	Request: OPTIONS sip:206@192.168.2.6:5060
6	0.245621	192.168.2.6	192.168.2.122	SIP/SDF	1036	Status: 200 OK

VoIP.pcap - VoIP Calls

Detected 2 VoIP Calls. Selected 1 Call.

Start Time	Stop Time	Initial Speaker	From	To	Protocol	Packets
0.244889	0.326367	192.168.2.122	"204" <sip:204@192.168.2.122>	<sip:206@192.168.2.6:5060>	SIP	1
100.981947	102.044162	192.168.1.21	<sip:kiakarz@clearsea.com "95505">	<sip:95505@clearsea.com>	SIP	1

Total: Calls: 2 Start packets: 0 Completed calls: 1 Rejected calls: 0

Prepare Filter Flow Player Select All

VoIP.pcap - VoIP - RTP Player

01:26:37 01:26:38 01:26:39 01:26:40 01:26:41 01:26:42 01:26:43 01:26:44

☒ From 192.168.2.122:10554 to 192.168.2.6:23000 Duration:10.68 Drop by Jitter Buff:532(59.8%) Out of Seq: 0(0.0%) Wrong Timestamp: 177(19.9%)

☐ From 192.168.2.122:16274 to 192.168.2.6:24000 Duration:0.00 Drop by Jitter Buff:204(87.6%) Out of Seq: 233(100.0%) Wrong Timestamp: 0(0.0%)

☒ From 192.168.2.6:24000 to 192.168.2.122:16274 Duration:0.03 Drop by Jitter Buff:262(73.8%) Out of Seq: 355(100.0%) Wrong Timestamp: 0(0.0%)

☐ From 192.168.2.6:23000 to 192.168.2.122:10554 Duration:10.80 Drop by Jitter Buff:540(59.9%) Out of Seq: 0(0.0%) Wrong Timestamp: 179(19.9%)

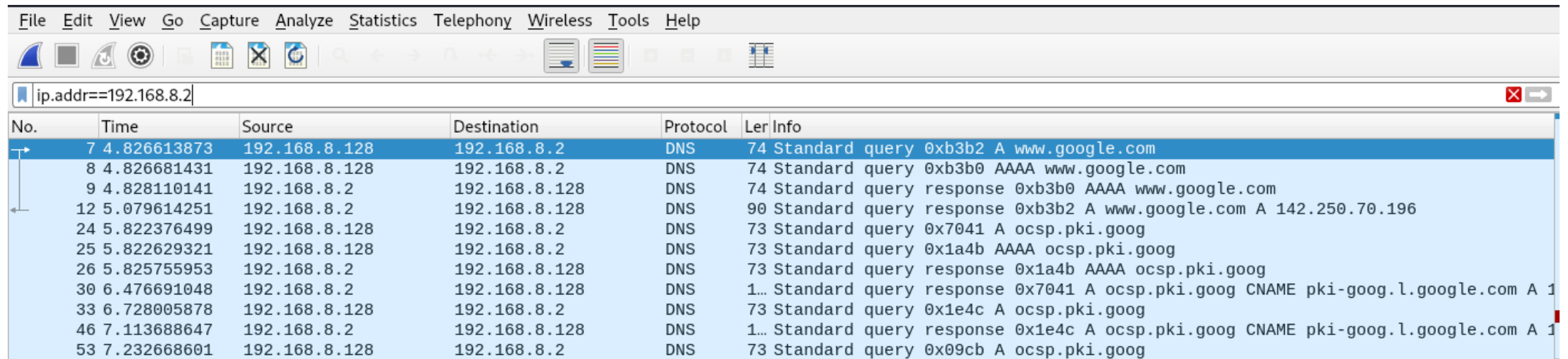
☒ View as time of day

Wireshark can filter packet

- Capture filter
 - Capture only **specific IP**
 - Capture only **specific protocol**
 - Capture only **specific port**
- Display filter
 - From all packet capture **display** the **wanted packets**

Wireshark packet filters

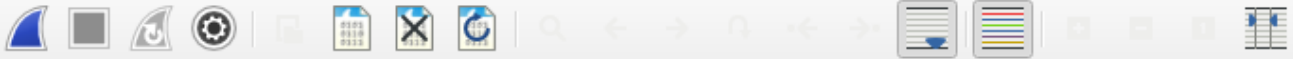
- Filter by IP address



The image shows the Wireshark network protocol analyzer interface. The top menu bar includes File, Edit, View, Go, Capture, Analyze, Statistics, Telephony, Wireless, Tools, and Help. Below the menu is a toolbar with various icons for file operations, capture control, and packet analysis. The packet filter bar at the top of the packet list pane contains the filter `ip.addr==192.168.8.2`. The packet list pane displays a table of captured packets, all of which are DNS packets. The first packet is a query from 192.168.8.128 to 192.168.8.2 for www.google.com. Subsequent packets show responses and further queries for the same domain and other subdomains like ojsp.pki.goog.

No.	Time	Source	Destination	Protocol	Len	Info
7	4.826613873	192.168.8.128	192.168.8.2	DNS	74	Standard query 0xb3b2 A www.google.com
8	4.826681431	192.168.8.128	192.168.8.2	DNS	74	Standard query 0xb3b0 AAAA www.google.com
9	4.828110141	192.168.8.2	192.168.8.128	DNS	74	Standard query response 0xb3b0 AAAA www.google.com
12	5.079614251	192.168.8.2	192.168.8.128	DNS	90	Standard query response 0xb3b2 A www.google.com A 142.250.70.196
24	5.822376499	192.168.8.128	192.168.8.2	DNS	73	Standard query 0x7041 A ojsp.pki.goog
25	5.822629321	192.168.8.128	192.168.8.2	DNS	73	Standard query 0x1a4b AAAA ojsp.pki.goog
26	5.825755953	192.168.8.2	192.168.8.128	DNS	73	Standard query response 0x1a4b AAAA ojsp.pki.goog
30	6.476691048	192.168.8.2	192.168.8.128	DNS	1...	Standard query response 0x7041 A ojsp.pki.goog CNAME pki-goog.l.google.com A 1
33	6.728005878	192.168.8.128	192.168.8.2	DNS	73	Standard query 0x1e4c A ojsp.pki.goog
46	7.113688647	192.168.8.2	192.168.8.128	DNS	1...	Standard query response 0x1e4c A ojsp.pki.goog CNAME pki-goog.l.google.com A 1
53	7.232668601	192.168.8.128	192.168.8.2	DNS	73	Standard query 0x09cb A ojsp.pki.goog

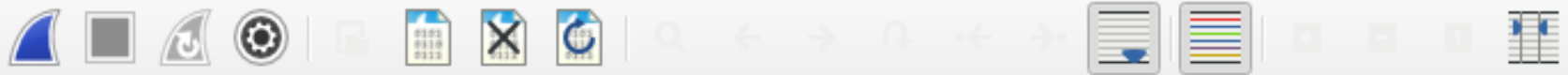
Filter by source host

File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help						
						
ip.src_host==192.168.8.128						
No.	Time	Source	Destination	Protocol	Len	Info
1	0.000000000	192.168.8.128	142.250.70.195	TCP	54	38302 → 80 [ACK] Seq=1 Ack=1 Win=63791 Len=0
3	1.310645955	192.168.8.128	35.244.181.201	TLSv1.2	1...	Application Data
6	1.611587158	192.168.8.128	35.244.181.201	TCP	54	36070 → 443 [ACK] Seq=47 Ack=47 Win=62780 Len=0
7	4.826613873	192.168.8.128	192.168.8.2	DNS	74	Standard query 0xb3b2 A www.google.com
8	4.826681431	192.168.8.128	192.168.8.2	DNS	74	Standard query 0xb3b0 AAAA www.google.com
10	4.867771182	192.168.8.128	152.195.38.76	TCP	54	55628 → 80 [ACK] Seq=1 Ack=1 Win=63503 Len=0
13	5.080076406	192.168.8.128	142.250.70.196	TCP	74	47448 → 443 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK
15	5.353521004	192.168.8.128	142.250.70.196	TCP	54	47448 → 443 [ACK] Seq=1 Ack=1 Win=64240 Len=0
16	5.355051435	192.168.8.128	142.250.70.196	TLSv1.3	5...	Client Hello
20	5.791373543	192.168.8.128	142.250.70.196	TCP	54	47448 → 443 [ACK] Seq=518 Ack=1461 Win=62780 Len=0
21	5.791521355	192.168.8.128	142.250.70.196	TCP	54	47448 → 443 [ACK] Seq=518 Ack=2921 Win=62780 Len=0

Filter by port number

File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help						
tcp.port == 80 udp.port == 80						
No.	Time	Source	Destination	Protocol	Len	Info
27	6.143694496	192.168.8.128	23.206.199.32	TCP	54	38268 → 80 [ACK] Seq=1 Ack=1 Win=64008 Len=0
28	6.144161910	23.206.199.32	192.168.8.128	TCP	60	[TCP ACKed unseen segment] 80 → 38268 [ACK] Seq=1 Ack=2 Win=64240 Len=0
31	6.477079526	192.168.8.128	142.250.70.227	TCP	74	40638 → 80 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM TSval=2854028878 TSe
38	6.791180302	142.250.70.227	192.168.8.128	TCP	60	80 → 40638 [SYN, ACK] Seq=0 Ack=1 Win=64240 Len=0 MSS=1460
39	6.791241509	192.168.8.128	142.250.70.227	TCP	54	40638 → 80 [ACK] Seq=1 Ack=1 Win=64240 Len=0
40	6.791682117	192.168.8.128	142.250.70.227	OCSP	4...	Request
41	6.791882724	142.250.70.227	192.168.8.128	TCP	60	80 → 40638 [ACK] Seq=1 Ack=419 Win=64240 Len=0
54	7.345807269	142.250.70.227	192.168.8.128	OCSP	7...	Response
55	7.345854442	192.168.8.128	142.250.70.227	TCP	54	40638 → 80 [ACK] Seq=419 Ack=702 Win=63791 Len=0
56	7.346303172	192.168.8.128	142.250.70.227	OCSP	4...	Request
57	7.346507869	142.250.70.227	192.168.8.128	TCP	60	80 → 40638 [ACK] Seq=702 Ack=837 Win=64240 Len=0

Filter by protocol

File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help						
						
http						
No.	Time	Source	Destination	Protocol	Len	Info
40	6.791682117	192.168.8.128	142.250.70.227	OCSP	4...	Request
54	7.345807269	142.250.70.227	192.168.8.128	OCSP	7...	Response
56	7.346303172	192.168.8.128	142.250.70.227	OCSP	4...	Request
75	8.198268551	142.250.70.227	192.168.8.128	OCSP	7...	Response
194	10.435076703	192.168.8.128	142.250.70.227	OCSP	4...	Request
224	11.025101152	142.250.70.227	192.168.8.128	OCSP	7...	Response
376	12.086320934	192.168.8.128	142.250.70.227	OCSP	4...	Request
452	12.200650050	192.168.8.128	142.250.70.195	OCSP	4...	Request
587	12.451265256	142.250.70.227	192.168.8.128	OCSP	7...	Response
589	12.452160634	192.168.8.128	142.250.70.227	OCSP	4...	Request
652	12.554166623	142.250.70.195	192.168.8.128	OCSP	7...	Response

Filter by multiple criteria

ip.addr == 192.168.8.128 and tcp.port == 80							 
.	Time	Source	Destination	Protocol	Len	Info	
1	0.000000000	192.168.8.128	142.250.70.195	TCP	54	38302 → 80 [ACK] Seq=1 Ack=1 Win=63791 Len=0	
2	0.000467688	142.250.70.195	192.168.8.128	TCP	60	[TCP ACKed unseen segment] 80 → 38302 [ACK] Seq=1 Ack=2 Win=64240 Len=0	
10	4.867771182	192.168.8.128	152.195.38.76	TCP	54	55628 → 80 [ACK] Seq=1 Ack=1 Win=63503 Len=0	
11	4.867920601	152.195.38.76	192.168.8.128	TCP	60	[TCP ACKed unseen segment] 80 → 55628 [ACK] Seq=1 Ack=2 Win=64240 Len=0	
27	6.143694496	192.168.8.128	23.206.199.32	TCP	54	38268 → 80 [ACK] Seq=1 Ack=1 Win=64008 Len=0	
28	6.144161910	23.206.199.32	192.168.8.128	TCP	60	[TCP ACKed unseen segment] 80 → 38268 [ACK] Seq=1 Ack=2 Win=64240 Len=0	
31	6.477079526	192.168.8.128	142.250.70.227	TCP	74	40638 → 80 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM TSval=2854028878 TSecr=	
38	6.791180302	142.250.70.227	192.168.8.128	TCP	60	80 → 40638 [SYN, ACK] Seq=0 Ack=1 Win=64240 Len=0 MSS=1460	
39	6.791241509	192.168.8.128	142.250.70.227	TCP	54	40638 → 80 [ACK] Seq=1 Ack=1 Win=64240 Len=0	
40	6.791682117	192.168.8.128	142.250.70.227	OCSP	4...	Request	
41	6.791882724	142.250.70.227	192.168.8.128	TCP	60	80 → 40638 [ACK] Seq=1 Ack=419 Win=64240 Len=0	

Capture username and password

The image shows a Wireshark packet capture interface. The main pane displays a list of network packets. Packet 171 is selected, and a detailed view pane on the right shows the contents of the selected packet, which is an HTTP POST request.

Packet List:

No.	Time	Source	Destination	Protocol	Len	Info
167	172.855469972	192.168.1.129	192.168.1.128	TCP	62	1056 → 80 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 S
168	172.855498311	192.168.1.128	192.168.1.129	TCP	62	80 → 1056 [SYN, ACK] Seq=0 Ack=1 Win=64240 Len=0
169	172.855752967	192.168.1.129	192.168.1.128	TCP	60	1056 → 80 [ACK] Seq=1 Ack=1 Win=64240 Len=0
170	172.856594538	192.168.1.129	192.168.1.128	TCP	60	[TCP Dup ACK 169#1] 1056 → 80 [ACK] Seq=1 Ack=1
171	172.856594538	192.168.1.129	192.168.1.128	HTTP	350	POST /device-based/regular/login/?login_attempt=1&lwv=100 HTTP/1.1
172	172.856594538	192.168.1.129	192.168.1.128	HTTP	350	Host: 192.168.1.128
173	172.856594538	192.168.1.129	192.168.1.128	HTTP	350	User-Agent: Mozilla/5.0 (Windows NT 5.1; rv:52.0) Gecko/20100101 Firefox/52.0
174	172.856594538	192.168.1.129	192.168.1.128	HTTP	350	Accept: text/html,application/xhtml+xml,application/xml;q=0.9,*/*;q=0.8
175	172.856594538	192.168.1.129	192.168.1.128	HTTP	350	Accept-Language: en-US,en;q=0.5
176	172.856594538	192.168.1.129	192.168.1.128	HTTP	350	Accept-Encoding: gzip, deflate
177	172.856594538	192.168.1.129	192.168.1.128	HTTP	350	Referer: http://192.168.1.128/

Packet 171 Details:

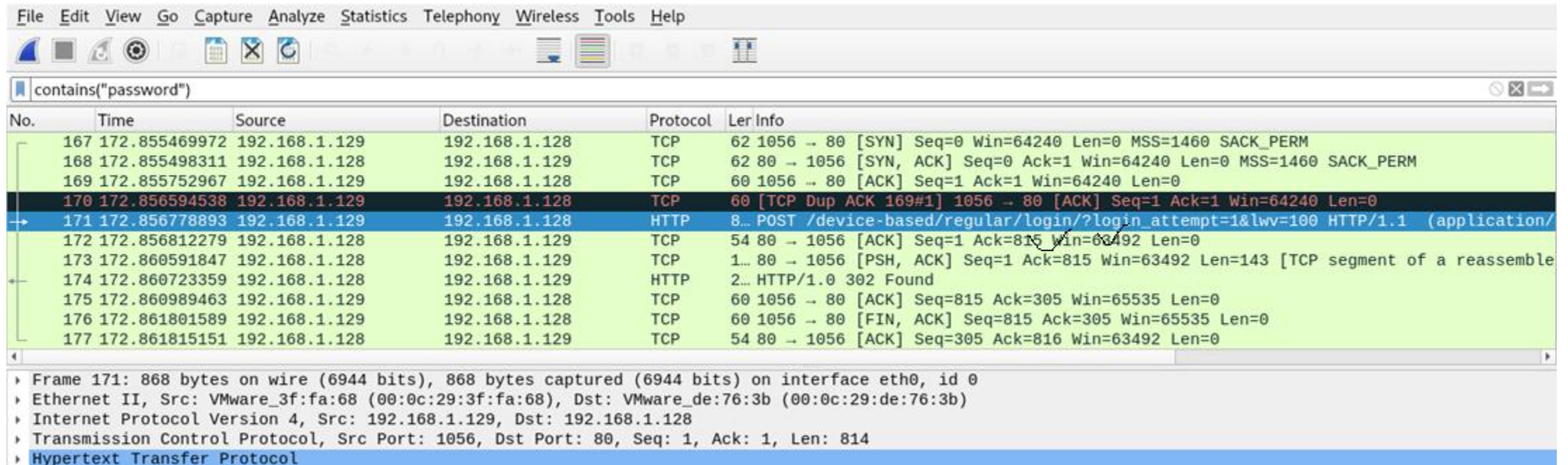
- Frame 171:
- Ethernet II
- Internet Protocol Version 4
- Transmission Control Protocol
- Hypertext Transfer Protocol
- HTML Form URL Encoded

HTTP POST Request Details:

```
POST /device-based/regular/login/?login_attempt=1&lwv=100 HTTP/1.1
Host: 192.168.1.128
User-Agent: Mozilla/5.0 (Windows NT 5.1; rv:52.0) Gecko/20100101 Firefox/52.0
Accept: text/html,application/xhtml+xml,application/xml;q=0.9,*/*;q=0.8
Accept-Language: en-US,en;q=0.5
Accept-Encoding: gzip, deflate
Referer: http://192.168.1.128/
Connection: keep-alive
Upgrade-Insecure-Requests: 1
Content-Type: application/x-www-form-urlencoded
Content-Length: 350

jazoest=2904&lsd=AVp8GMaFiPU&display=&isprivate=&return_session=&skip_api_login=&signed_next=&trynum=1&
timezone=&lgnrdim=&lgnrnd=060213_4h2w&lgnjs=n&email=abc%40gmail.com&pass=123456&login=1&prefill_contact_
point=&prefill_source=&prefill_type=&first_prefill_source=&first_prefill_type=&had_cp_prefilled=false&h
ad_password_prefilled=false&ab_test_data=HTTP/1.0 302 Found
Server: BaseHTTP/0.6 Python/3.11.6
Date: Tue, 02 Jan 2024 14:20:28 GMT
Location: https://login.facebook.com/login.php
```


Contain filter



The image shows a Wireshark packet capture window with a filter applied: `contains("password")`. The packet list shows several packets, with packet 171 selected. The packet details pane shows the structure of the selected packet, which is an HTTP POST request.

No.	Time	Source	Destination	Protocol	Len	Info
167	172.855469972	192.168.1.129	192.168.1.128	TCP	62	1056 → 80 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM
168	172.855498311	192.168.1.128	192.168.1.129	TCP	62	80 → 1056 [SYN, ACK] Seq=0 Ack=1 Win=64240 Len=0 MSS=1460 SACK_PERM
169	172.855752967	192.168.1.129	192.168.1.128	TCP	60	1056 → 80 [ACK] Seq=1 Ack=1 Win=64240 Len=0
170	172.856594538	192.168.1.129	192.168.1.128	TCP	60	[TCP Dup ACK 169#1] 1056 → 80 [ACK] Seq=1 Ack=1 Win=64240 Len=0
171	172.856778893	192.168.1.129	192.168.1.128	HTTP	8...	POST /device-based/regular/login/?login_attempt=1&lwv=100 HTTP/1.1 (application/
172	172.856812279	192.168.1.128	192.168.1.129	TCP	54	80 → 1056 [ACK] Seq=1 Ack=815 Win=63492 Len=0
173	172.860591847	192.168.1.128	192.168.1.129	TCP	1...	80 → 1056 [PSH, ACK] Seq=1 Ack=815 Win=63492 Len=143 [TCP segment of a reassembled
174	172.860723359	192.168.1.128	192.168.1.129	HTTP	2...	HTTP/1.0 302 Found
175	172.860989463	192.168.1.129	192.168.1.128	TCP	60	1056 → 80 [ACK] Seq=815 Ack=305 Win=65535 Len=0
176	172.861801589	192.168.1.129	192.168.1.128	TCP	60	1056 → 80 [FIN, ACK] Seq=815 Ack=305 Win=65535 Len=0
177	172.861815151	192.168.1.129	192.168.1.128	TCP	54	80 → 1056 [ACK] Seq=305 Ack=816 Win=63492 Len=0

Frame 171: 868 bytes on wire (6944 bits), 868 bytes captured (6944 bits) on interface eth0, id 0
Ethernet II, Src: VMware_3f:fa:68 (00:0c:29:3f:fa:68), Dst: VMware_de:76:3b (00:0c:29:de:76:3b)
Internet Protocol Version 4, Src: 192.168.1.129, Dst: 192.168.1.128
Transmission Control Protocol, Src Port: 1056, Dst Port: 80, Seq: 1, Ack: 1, Len: 814
Hypertext Transfer Protocol

http.request.method == "POST"

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http.request.method == "POST"

No.	Time	Source	Destination	Protocol	Len	Info
171	172.856778893	192.168.1.129	192.168.1.128	HTTP	814	POST /device-based/...<div data-bbox="655 405 875 900" data-label="List-Group"> - Mark/Unmark Packet Ctrl+M - Ignore/Unignore Packet Ctrl+D - Set/Unset Time Reference Ctrl+T - Time Shift... Ctrl+Shift+T - Packet Comments - Edit Resolved Name - Apply as Filter - Prepare as Filter - Conversation Filter - Colorize Conversation - SCTP - Follow ✓ - Copy - Protocol Preferences - Decode As... - Show Packet in New Window

http password.pcapng

File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help

http

Source	Destination	Protocol	Length	Info
192.168.0.100	224.0.0.251	MDNS	152	Standard query 0x0041 PTR _%9E5E7C8F47989526C9BCD95D24084F6F0B27C
192.168.0.104	224.0.0.251	MDNS	437	Standard query response 0x0000 PTR Y-Series-896e8e3a4462f9a459a9f
192.168.0.104	224.0.0.251	MDNS	404	Standard query response 0x0000 PTR Y-Series-896e8e3a4462f9a459a9f
192.168.0.104	224.0.0.251	MDNS	389	Standard query response 0x0000 PTR Y-Series-896e8e3a4462f9a459a9f
192.168.0.1	224.0.0.1	IGMPv3	60	Membership Query, general
192.168.0.107	18.192.172.30	TCP	74	52356 → 80 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM=1 TSval=
18.192.172.30	192.168.0.107	TCP	74	80 → 52356 [SYN, ACK] Seq=0 Ack=1 Win=28960 Len=0 MSS=1440 SACK_F
192.168.0.107	18.192.172.30	TCP	66	52356 → 80 [ACK] Seq=1 Ack=1 Win=64256 Len=0 TSval=2163476701 TSe
192.168.0.107	18.192.172.30	HTTP	605	POST /userinfo.php HTTP/1.1 (application/x-www-form-urlencoded)
18.192.172.30	192.168.0.107	TCP	66	80 → 52356 [ACK] Seq=1 Ack=540 Win=30208 Len=0 TSval=773377626 TS
18.192.172.30	192.168.0.107	HTTP	342	HTTP/1.1 302 Found (text/html)
192.168.0.107	18.192.172.30	TCP	66	52356 → 80 [ACK] Seq=540 Ack=277 Win=64128 Len=0 TSval=2163476852
192.168.0.107	18.192.172.30	HTTP	460	GET /login.php HTTP/1.1

Wireshark · Packet 402 · http password.pcapng

Accept-Language: en-US,en;q=0.5\r\n
Accept-Encoding: gzip, deflate\r\n
Content-Type: application/x-www-form-urlencoded\r\n
> Content-Length: 36\r\n
Origin: http://testphp.vulnweb.com\r\n
Connection: keep-alive\r\n
Referer: http://testphp.vulnweb.com/login.php\r\n
Upgrade-Insecure-Requests: 1\r\n
\r\n
[Full request URI: http://testphp.vulnweb.com/userinfo.php]
[HTTP request 1/2]
[Response in frame: 404]
[Next request in frame: 406]
File Data: 36 bytes

HTML Form URL Encoded: application/x-www-form-urlencoded

- Form item: "uname" = "vijaymehta"
 - Key: uname
 - Value: vijaymehta
- Form item: "pass" = "maxelladiviner"
 - Key: pass
 - Value: maxelladiviner

Statistics packet

DDOS_Attack.pcap [Wireshark 1.10.2 (SVN Rev 51934 from /trunk-1.10)]

File Edit View Go Capture Analyze **Statistics** Telephony Tools Internals Help

Filter:

No.	Time	Source
1	0.000000	151.152.179.2
2	0.000198	209.73.12.194
3	0.000384	234.195.124.4
4	0.000385	24.5.195.21
5	0.000386	209.248.31.18
6	0.000472	209.73.12.194
7	0.000524	209.73.12.194
8	0.001014	195.236.255.2
9	0.001094	209.73.12.194
10	0.005320	205.207.103.6
11	0.005401	209.73.12.194
12	0.005580	176.225.20.4
13	0.005664	209.73.12.194
14	0.006010	27.87.212.69
15	0.006012	131.44.134.84

Statistics menu options:

- Summary
 - Comments Summary
 - Show address resolution
 - Protocol Hierarchy
- Conversations
- Endpoints
 - Packet Lengths...
- IO Graph
 - Conversation List
 - Endpoint List
 - Service Response Time
- ANCP
- BACnet
- BOOTP-DHCP...
- Collectd...
- Compare...
- Flow Graph...
- HART-IP
- HTTP
- ONC-RPC Programs
- Sametime
- TCP StreamGraph
- UDP Multicast Streams
- WLAN Traffic
- IP Destinations
- IP Addresses
- IP Protocol Types

Packet details:

- Frame 1: 174 bytes on wire (C)
- Ethernet II, Src: IntelCor_c
- Internet Protocol Version 4,
- Transmission Control Protocol

Multi devices sent one device

DDOS_Attack.pcap [Wireshark 1.10.2 (SVN Rev 51934 from /trunk-1.10)]

File Edit View Go Capture Analyze Statistics Telephony Tools Internals Help

Filter: Expression... Clear Apply Save

No.	Time	Source	Destination
1	0.000000	151.152.179.248	209.73.12.194
2	0.000198	209.73.12.194	151.152.179.248
3	0.000384	234.195.124.44	209.73.12.194
4	0.000385	24.5.195.21	209.73.12.194
5	0.000386	209.248.31.184	209.73.12.194
6	0.000472	209.73.12.194	24.5.195.21
7	0.000524	209.73.12.194	209.248.31.184
8	0.001014	195.236.255.27	209.73.12.194
9	0.001094	209.73.12.194	195.236.255.27
10	0.005320	205.207.103.69	209.73.12.194
11	0.005401	209.73.12.194	205.207.103.69
12	0.005580	176.225.20.4	209.73.12.194
13	0.005664	209.73.12.194	176.225.20.4
14	0.006010	27.87.212.69	209.73.12.194
15	0.006012	121.44.134.81	209.73.12.194

Frame 1: 174 bytes on wire (1392 bits), 174 bytes captured (1392 bits) on interface 0
Ethernet II, Src: IntelCor_cc:6f:16 (00:21:6b:cc:6f:16), Dst: 08:00:27:87:21:26 (08:00:27:87:21:26)
Internet Protocol Version 4, Src: 151.152.179.248, Dst: 209.73.12.194
Transmission Control Protocol, Src Port: firefox (54012), Dst Port: 80

Conversations: DDOS_Attack.pcap

Ethernet: 2 Fibre Channel FDDI IPv4: 4594 IPv6 IPX JXTA NCP RSVP SCTP TCP: 4594 Token Ring UDP USB WLAN

IPv4 Conversations

Address A	Address B	Packets	Bytes	Packets A→B	Bytes A→B	Packets B→A	Bytes B→A	Rel Start	Durati
151.152.179.248	209.73.12.194	3	290	1	174	2	116	0.000000000	
209.73.12.194	234.195.124.44	1	174	0	0	1	174	0.000384000	
24.5.195.21	209.73.12.194	3	290	1	174	2	116	0.000385000	
209.73.12.194	209.248.31.184	3	290	2	116	1	174	0.000386000	
195.236.255.27	209.73.12.194	3	290	1	174	2	116	0.001014000	
205.207.103.69	209.73.12.194	3	290	1	174	2	116	0.005320000	
176.225.20.4	209.73.12.194	3	290	1	174	2	116	0.005580000	
27.87.212.69	209.73.12.194	3	290	1	174	2	116	0.006010000	
121.44.134.81	209.73.12.194	3	290	1	174	2	116	0.006012000	
35.43.176.25	209.73.12.194	3	290	1	174	2	116	0.006012000	
209.73.12.194	242.113.234.146	1	174	0	0	1	174	0.006013000	
70.14.207.92	209.73.12.194	3	290	1	174	2	116	0.006013000	
90.55.27.112	209.73.12.194	3	290	1	174	2	116	0.006455000	
177.126.227.163	209.73.12.194	3	290	1	174	2	116	0.006456000	
209.73.12.194	237.41.70.230	1	174	0	0	1	174	0.006457000	
121.236.239.184	209.73.12.194	3	290	1	174	2	116	0.006457000	
209.73.12.194	254.1.72.28	1	174	0	0	1	174	0.007562000	
209.73.12.194	248.182.253.29	1	174	0	0	1	174	0.007764000	
31.163.59.109	209.73.12.194	3	290	1	174	2	116	0.007914000	
110.105.41.189	209.73.12.194	3	290	1	174	2	116	0.007915000	
200.236.236.45	209.73.12.194	3	290	1	174	2	116	0.009956000	

☒ Name resolution ☐ Limit to display filter

Help Copy Follow Stream Graph A→B Graph B→A Close

All
devices
talk on
the same
port

DDOS_Attack.pcap [Wireshark 1.10.2 (SVN Rev 51934 from /trunk-1.10)]

File Edit View Go Capture Analyze Statistics Telephony Tools Internals Help

Filter: Expression... Clear Apply Save

No.	Time	Source	Destination
1	0.000000	151.152.179.248	209.73.12.194
2	0.000198	209.73.12.194	151.152.179.248
3	0.000384	234.195.124.44	209.73.12.194
4	0.000385	24.5.195.21	209.73.12.194
5	0.000386	209.248.31.184	209.73.12.194
6	0.000472	209.73.12.194	24.5.195.21
7	0.000524	209.73.12.194	209.248.31.184
8	0.001014	195.236.255.27	209.73.12.194
9	0.001094	209.73.12.194	195.236.255.27
10	0.005320	205.207.103.69	209.73.12.194
11	0.005401	209.73.12.194	205.207.103.69
12	0.005580	176.225.20.4	209.73.12.194
13	0.005664	209.73.12.194	176.225.20.4
14	0.006010	27.87.212.69	209.73.12.194
15	0.006012	121.44.134.81	209.73.12.194

Frame 1: 174 bytes on wire (1392 bits), 174 bytes captured (1392 bits) on interface 0
Ethernet II, Src: IntelCor_cc:6f:16 (00:21:6b:cc:6f:16), Dst: 08:00:27:12:19:40 (08:00:27:12:19:40)
Internet Protocol Version 4, Src: 151.152.179.248, Dst: 209.73.12.194
Transmission Control Protocol, Src Port: firefox (1511), Dst Port: http (80)

Conversations: DDOS_Attack.pcap

Ethernet: 2 Fibre Channel FDDI IPv4: 4594 IPv6 IPX JXTA NCP RSVP SCTP TCP: 4594 Token Ring UDP USB WLAN

TCP Conversations

Address A	Port A	Address B	Port B	Packets	Bytes	Packets A→B	Bytes A→B	Packets B→A	Bytes B→A
151.152.179.248	firefox	209.73.12.194	http	3	290	1	174	2	116
234.195.124.44	ng-umds	209.73.12.194	http	1	174	1	174	0	0
24.5.195.21	empire-e	209.73.12.194	http	3	290	1	174	2	116
209.248.31.184	sstsys-lm	209.73.12.194	http	3	290	1	174	2	116
195.236.255.27	rritr	209.73.12.194	http	3	290	1	174	2	116
205.207.103.69	rimwm	209.73.12.194	http	3	290	1	174	2	116
176.225.20.4	rilwm	209.73.12.194	http	3	290	1	174	2	116
27.87.212.69	rrifmm	209.73.12.194	http	3	290	1	174	2	116
121.44.134.81	rrisat	209.73.12.194	http	3	290	1	174	2	116
35.43.176.25	rsyp-enci	209.73.12.194	http	3	290	1	174	2	116
242.113.234.146	rsyp-enci	209.73.12.194	http	1	174	1	174	0	0
70.14.207.92	mps-raft	209.73.12.194	http	3	290	1	174	2	116
90.55.27.112	l2f	209.73.12.194	http	3	290	1	174	2	116
177.126.227.163	deskshart	209.73.12.194	http	3	290	1	174	2	116
237.41.70.230	hb-engin	209.73.12.194	http	1	174	1	174	0	0
121.236.239.184	bcs-broki	209.73.12.194	http	3	290	1	174	2	116
254.1.72.28	slingshot	209.73.12.194	http	1	174	1	174	0	0
248.182.253.29	jetform	209.73.12.194	http	1	174	1	174	0	0
31.163.59.109	vdmply	209.73.12.194	http	3	290	1	174	2	116
110.105.41.189	gat-lmd	209.73.12.194	http	3	290	1	174	2	116
200.236.236.45	centra	209.73.12.194	http	3	290	1	174	2	116

☒ Name resolution ☐ Limit to display filter

Help Copy Follow Stream Graph A→B Graph B→A Close

Decrypt file data

- Open HTTPS/TLS.pcapng
- The **3-way handshake** is happening
- Hello from **SSL Client** and the **ACK** from server
- **Server Hello** and then **ACK**
- **Exchanging** some key and Cipher information
- Started **Exchanging Data**.

No.	Time	Source	Destination	Protocol	Length	Info
1	0.000000	127.0.0.1	127.0.0.1	TCP	74	38713 → 443 [SYN] Seq=0 Win=32767 Len=0 MSS=16396 SAC
2	0.000021	127.0.0.1	127.0.0.1	TCP	74	443 → 38713 [SYN, ACK] Seq=0 Ack=1 Win=32767 Len=0 MS
3	0.000037	127.0.0.1	127.0.0.1	TCP	66	38713 → 443 [ACK] Seq=1 Ack=1 Win=32767 Len=0 TSval=5
4	0.000158	127.0.0.1	127.0.0.1	SSLv2	171	Client Hello ←
5	0.000178	127.0.0.1	127.0.0.1	TCP	66	443 → 38713 [ACK] Seq=1 Ack=106 Win=32767 Len=0 TSval
6	0.002160	127.0.0.1	127.0.0.1	SSLv3	995	Server Hello, Certificate, Server Hello Done ←
7	0.002609	127.0.0.1	127.0.0.1	TCP	66	38713 → 443 [ACK] Seq=106 Ack=930 Win=32767 Len=0 TSv
8	2.808933	127.0.0.1	127.0.0.1	SSLv3	278	Client Key Exchange, Change Cipher Spec, Encrypted Ha
9	2.822770	127.0.0.1	127.0.0.1	SSLv3	141	Change Cipher Spec, Encrypted Handshake Message
10	2.822809	127.0.0.1	127.0.0.1	TCP	66	38713 → 443 [ACK] Seq=318 Ack=1005 Win=32767 Len=0 TS
11	2.833071	127.0.0.1	127.0.0.1	SSLv3	503	Application Data ←
12	2.873275	127.0.0.1	127.0.0.1	TCP	66	443 → 38713 [ACK] Seq=1005 Ack=755 Win=32767 Len=0 TS
13	2.938485	127.0.0.1	127.0.0.1	SSLv3	103	Encrypted Handshake Message
14	2.938750	127.0.0.1	127.0.0.1	SSLv3	183	Encrypted Handshake Message
15	2.938761	127.0.0.1	127.0.0.1	TCP	66	443 → 38713 [ACK] Seq=1042 Ack=872 Win=32767 Len=0 TS
16	2.938999	127.0.0.1	127.0.0.1	SSLv3	1073	Encrypted Handshake Message, Encrypted Handshake Mess
17	2.940026	127.0.0.1	127.0.0.1	SSLv3	337	Encrypted Handshake Message, Change Cipher Spec, Encr
18	2.943406	127.0.0.1	127.0.0.1	SSLv3	172	Change Cipher Spec, Encrypted Handshake Message
19	2.944825	127.0.0.1	127.0.0.1	SSLv3	5756	Application Data, Application Data

> Frame 11: 503 bytes on wire (4024 bits), 503 bytes captured (4024 bits)
> Ethernet II, Src: 00:00:00_00:00:00 (00:00:00:00:00:00), Dst: 00:00:00_00:00:00 (00:00:00:00:00:00)
> Internet Protocol Version 4, Src: 127.0.0.1, Dst: 127.0.0.1
> Transmission Control Protocol, Src Port: 38713, Dst Port: 443, Seq: 318, Ack: 1005, Len: 437
> Transport Layer Security

Decrypt file data

- Then, if we click on any **application data**, that **data is unreadable** to us.
- However, with Wireshark, we can **decrypt that data...** all we need is the **server's Private Key**.
- ***Don't worry we have already provided the key along with the PCAP file.***

No.	Time	Source	Destination	Protocol	Length	Info
1	0.000000	127.0.0.1	127.0.0.1	TCP	74	38713 → 443 [SYN] Seq=0 Win=32767 Len=4
2	0.000021	127.0.0.1	127.0.0.1	TCP	74	443 → 38713 [SYN, ACK] Seq=0 Ack=1 Win=
3	0.000037	127.0.0.1	127.0.0.1	TCP	66	38713 → 443 [ACK] Seq=1 Ack=1 Win=3276
4	0.000158	127.0.0.1	127.0.0.1	SSLv2	171	Client Hello
5	0.000178	127.0.0.1	127.0.0.1	TCP	66	443 → 38713 [ACK] Seq=1 Ack=106 Win=32
6	0.002160	127.0.0.1	127.0.0.1	SSLv3	995	Server Hello, Certificate, Server Hello
7	0.002609	127.0.0.1	127.0.0.1	TCP	66	38713 → 443 [ACK] Seq=106 Ack=930 Win=
8	2.808933	127.0.0.1	127.0.0.1	SSLv3	278	Client Key Exchange, Change Cipher Spe
9	2.822770	127.0.0.1	127.0.0.1	SSLv3	141	Change Cipher Spec, Encrypted Handshake
10	2.822809	127.0.0.1	127.0.0.1	TCP	66	38713 → 443 [ACK] Seq=318 Ack=1005 Win=
11	2.833071	127.0.0.1	127.0.0.1	SSLv3	503	Application Data
12	2.873275	127.0.0.1	127.0.0.1	TCP	66	443 → 38713 [ACK] Seq=1005 Ack=755 Win=
13	2.938485	127.0.0.1	127.0.0.1	SSLv3	103	Encrypted Handshake Message
14	2.938750	127.0.0.1	127.0.0.1	SSLv3	183	Encrypted Handshake Message
15	2.938761	127.0.0.1	127.0.0.1	TCP	66	443 → 38713 [ACK] Seq=1042 Ack=872 Win=
16	2.938999	127.0.0.1	127.0.0.1	SSLv3	1073	Encrypted Handshake Message, Encrypted
17	2.940026	127.0.0.1	127.0.0.1	SSLv3	337	Encrypted Handshake Message, Change Ci
18	2.943406	127.0.0.1	127.0.0.1	SSLv3	172	Change Cipher Spec, Encrypted Handshake
19	2.944825	127.0.0.1	127.0.0.1	SSLv3	5756	Application Data, Application Data

<

> Frame 11: 503 bytes on wire (4024 bits), 503 bytes captured (4024 bits)

> Ethernet II, Src: 00:00:00:00:00:00 (00:00:00:00:00:00), Dst: 00:00:00:00:00:00 (00:00:00:00:00:00)

> Internet Protocol Version 4, Src: 127.0.0.1, Dst: 127.0.0.1

> Transmission Control Protocol, Src Port: 38713, Dst Port: 443, Seq: 318, Ack: 1005, Len: 437

▼ Transport Layer Security

 ▼ SSLv3 Record Layer: Application Data Protocol: Application Data

 Content Type: Application Data (23)

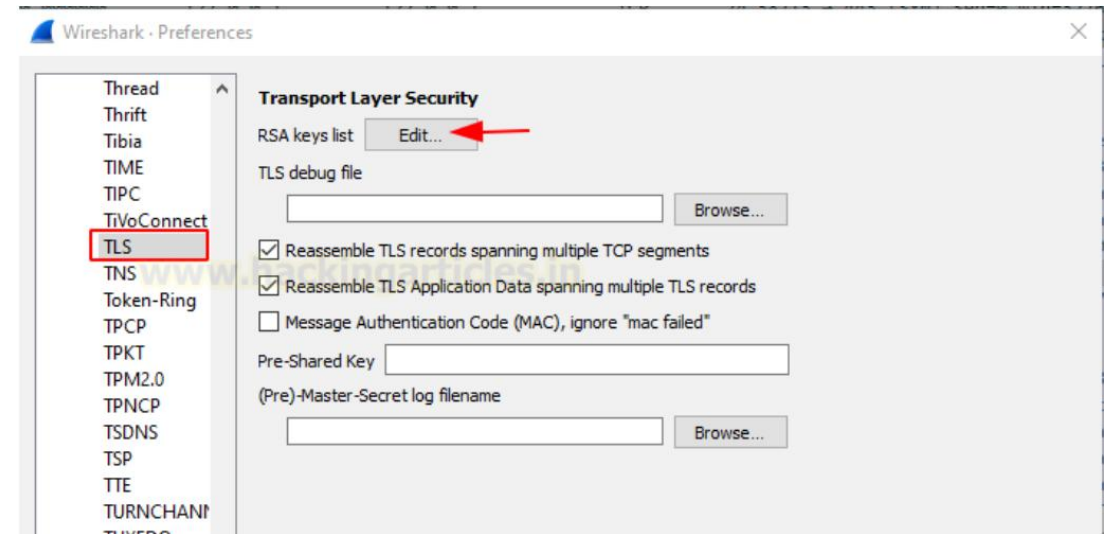
 Version: SSL 3.0 (0x0300)

 Length: 432

 Encrypted Application Data: 4ac33e9d7778012cb4bc4c9a84d7b9900c2110f0fa007c16bb77fb72424fad504ad0aa6f...

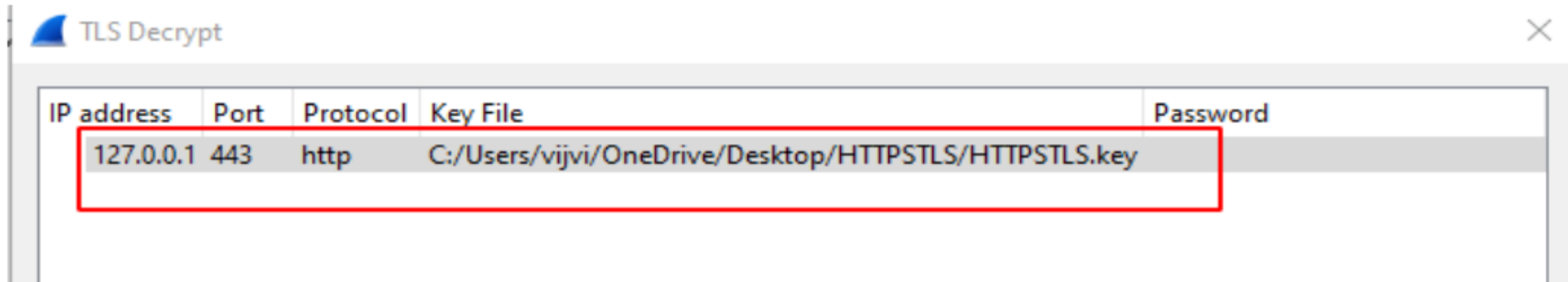
Decrypt data

- To **Decrypt the Encrypted Application Data** over TLS or SSL
Navigate to
- ***Edit > Preference > Protocol > TLS***



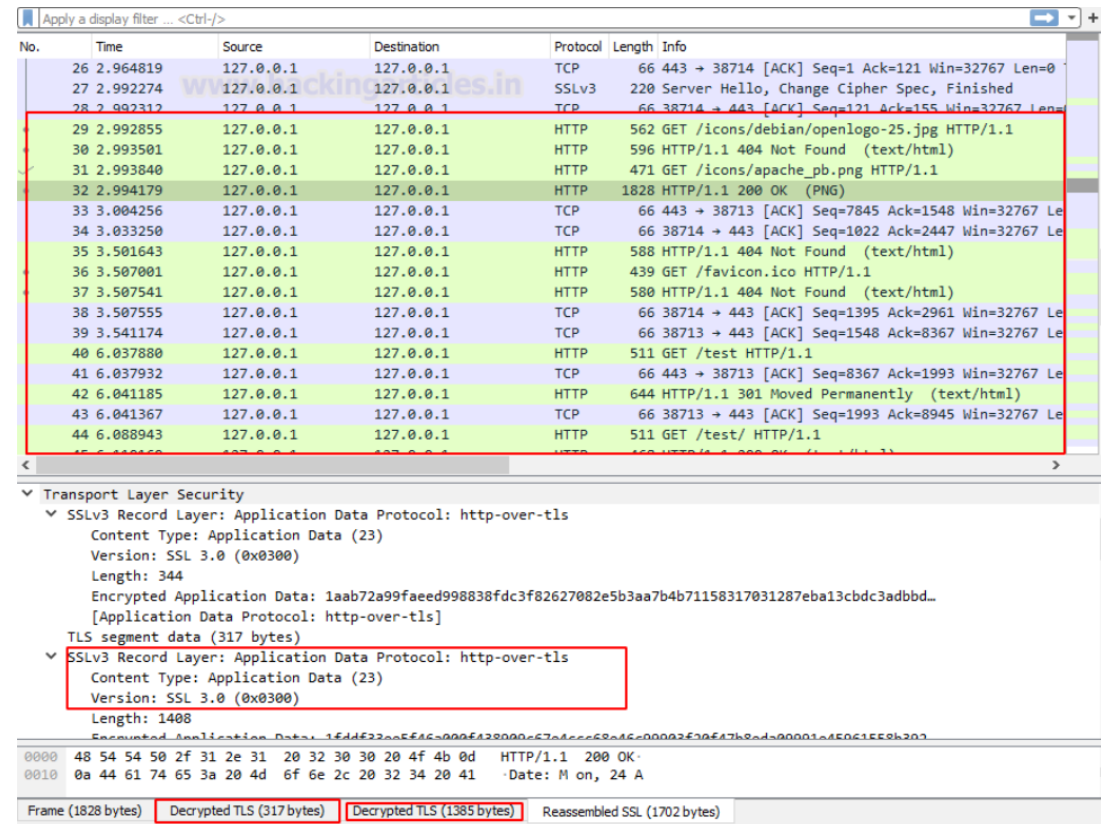
Add the path for file key

- And add these values
- **IP address: 127.0.0.1**
- **Port: 443**
- **Key File:**



Decrypted data

- Hurray!!! As you can see, we have Successfully decrypted the Data over the TLS



Apply a display filter ... <Ctrl-/>

No.	Time	Source	Destination	Protocol	Length	Info
26	2.964819	127.0.0.1	127.0.0.1	TCP	66	443 → 38714 [ACK] Seq=1 Ack=121 Win=32767 Len=0
27	2.992274	127.0.0.1	127.0.0.1	SSLv3	220	Server Hello, Change Cipher Spec, Finished
28	2.992312	127.0.0.1	127.0.0.1	TCP	66	38714 → 443 [ACK] Seq=121 Ack=155 Win=32767 Len=0
29	2.992855	127.0.0.1	127.0.0.1	HTTP	562	GET /icons/debian/openlogo-25.jpg HTTP/1.1
30	2.993501	127.0.0.1	127.0.0.1	HTTP	596	HTTP/1.1 404 Not Found (text/html)
31	2.993840	127.0.0.1	127.0.0.1	HTTP	471	GET /icons/apache_pb.png HTTP/1.1
32	2.994179	127.0.0.1	127.0.0.1	HTTP	1828	HTTP/1.1 200 OK (PNG)
33	3.004256	127.0.0.1	127.0.0.1	TCP	66	443 → 38713 [ACK] Seq=7845 Ack=1548 Win=32767 Len=0
34	3.033250	127.0.0.1	127.0.0.1	TCP	66	38714 → 443 [ACK] Seq=1022 Ack=2447 Win=32767 Len=0
35	3.501643	127.0.0.1	127.0.0.1	HTTP	588	HTTP/1.1 404 Not Found (text/html)
36	3.507001	127.0.0.1	127.0.0.1	HTTP	439	GET /favicon.ico HTTP/1.1
37	3.507541	127.0.0.1	127.0.0.1	HTTP	580	HTTP/1.1 404 Not Found (text/html)
38	3.507555	127.0.0.1	127.0.0.1	TCP	66	38714 → 443 [ACK] Seq=1395 Ack=2961 Win=32767 Len=0
39	3.541174	127.0.0.1	127.0.0.1	TCP	66	38713 → 443 [ACK] Seq=1548 Ack=8367 Win=32767 Len=0
40	6.037880	127.0.0.1	127.0.0.1	HTTP	511	GET /test HTTP/1.1
41	6.037932	127.0.0.1	127.0.0.1	TCP	66	443 → 38713 [ACK] Seq=8367 Ack=1993 Win=32767 Len=0
42	6.041185	127.0.0.1	127.0.0.1	HTTP	644	HTTP/1.1 301 Moved Permanently (text/html)
43	6.041367	127.0.0.1	127.0.0.1	TCP	66	38713 → 443 [ACK] Seq=1993 Ack=8945 Win=32767 Len=0
44	6.088943	127.0.0.1	127.0.0.1	HTTP	511	GET /test/ HTTP/1.1

Transport Layer Security

- SSLv3 Record Layer: Application Data Protocol: http-over-tls
 - Content Type: Application Data (23)
 - Version: SSL 3.0 (0x0300)
 - Length: 344
 - Encrypted Application Data: 1aab72a99faeed998838fdc3f82627082e5b3aa7b4b71158317031287eba13cbdc3adbbd...
 - [Application Data Protocol: http-over-tls]
 - TLS segment data (317 bytes)
- SSLv3 Record Layer: Application Data Protocol: http-over-tls
 - Content Type: Application Data (23)
 - Version: SSL 3.0 (0x0300)
 - Length: 1408
 - Encrypted Application Data: 1fddfe23ae5f46e000f432000c67a4e6658a46e00002f30ef47b8ad00001a45061f558b303...

0000 48 54 54 50 2f 31 2e 31 20 32 30 30 20 4f 4b 0d HTTP/1.1 200 OK
0010 0a 44 61 74 65 3a 20 4d 6f 6e 2c 20 32 34 20 41 Date: Mon, 24 A

Frame (1828 bytes) Decrypted TLS (317 bytes) Decrypted TLS (1385 bytes) Reassembled SSL (1702 bytes)

Tools used with Wireshark

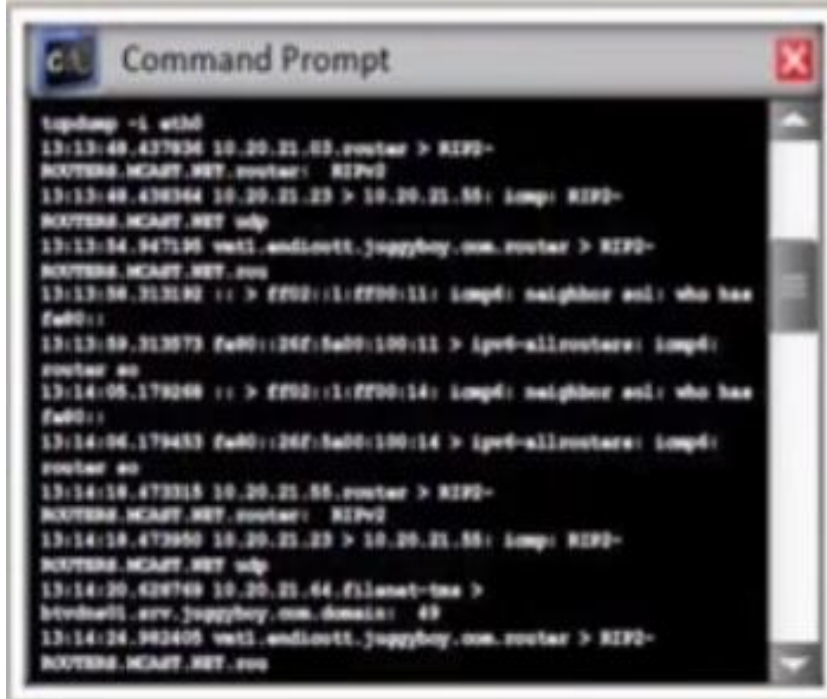
- Steel Central packet analyzer: give dashboard for packet captures in Wireshark
- Riverbed Packet Analyzer



Sniffing tools

TCPDump

Runs on Linux and UNIX systems



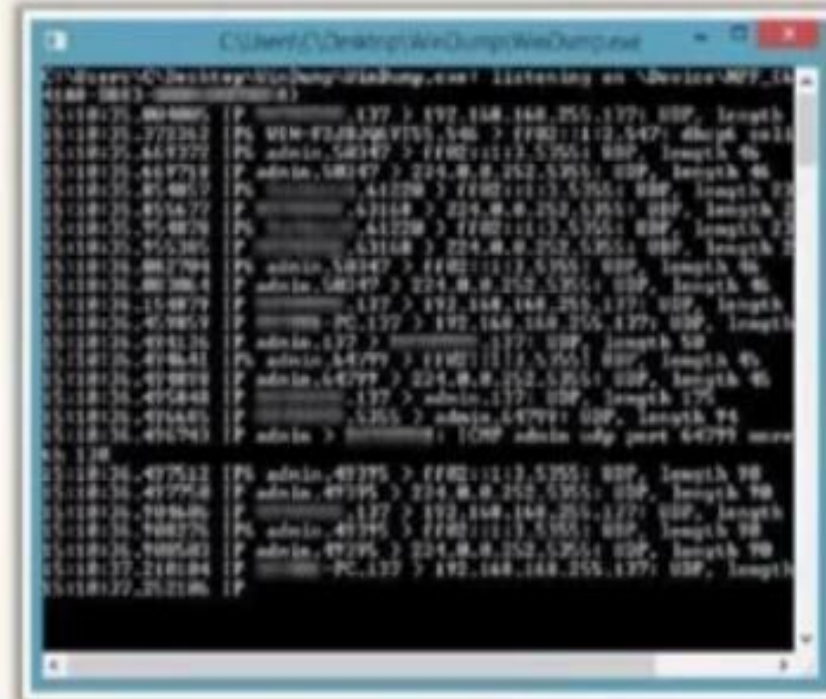
```
Command Prompt

tcpdump -i eth0
13:13:40.437834 10.20.21.83.router > RIPv2-
NEXTHop: MCAST.NET.router: RIPv2
13:13:40.438344 10.20.21.23 > 10.20.21.55: icmp: RIPv2-
NEXTHop: MCAST.NET udp
13:13:54.947196 vett1.andiottt.jaggyboy.com.router > RIPv2-
NEXTHop: MCAST.NET.ew
13:13:56.313192 :: > ff02::1:ff00:11: icmp4: neighbor sol: who has
fe80::
13:13:59.313073 fe80::26f:fe00:100:11 > ipv6-allrouters: icmp4:
router so
13:14:05.179248 :: > ff02::1:ff00:14: icmp4: neighbor sol: who has
fe80::
13:14:06.179453 fe80::26f:fe00:100:14 > ipv6-allrouters: icmp4:
router so
13:14:19.473315 10.20.21.55.router > RIPv2-
NEXTHop: MCAST.NET.router: RIPv2
13:14:19.473850 10.20.21.23 > 10.20.21.55: icmp: RIPv2-
NEXTHop: MCAST.NET udp
13:14:20.628749 10.20.21.64.filenet-tx >
btvdcall.srv.jaggyboy.com.domain: 49
13:14:24.982405 vett1.andiottt.jaggyboy.com.router > RIPv2-
NEXTHop: MCAST.NET.ew
```

<http://www.tcpdump.org>

WinDump

Runs on Windows systems



```
C:\Users\...> C:\Program Files\WinDump\WinDump.exe
C:\Users\...> C:\Program Files\WinDump\WinDump.exe listening on \Device\NPF{...}
05:18:13.000000 P > 192.168.168.255:137: UDP, length
05:18:13.272362 P WIN-473850VTS5.540 > ff02::1:2:547: dhcp call
05:18:13.448772 P admin.58347 > ff02::1:3:5355: UDP, length 46
05:18:13.469718 P admin.58347 > 224.0.0.252:5355: UDP, length 46
05:18:13.470517 P > ff02::1:3:5355: UDP, length 23
05:18:13.47148 > 224.0.0.252:5355: UDP, length 2
05:18:13.47148 > ff02::1:3:5355: UDP, length 23
05:18:13.47148 > 224.0.0.252:5355: UDP, length 2
05:18:13.47148 > ff02::1:3:5355: UDP, length 46
05:18:13.47148 > 224.0.0.252:5355: UDP, length 46
05:18:13.47148 > 192.168.168.255:137: UDP, length
05:18:13.47148 > 192.168.168.255:137: UDP, length
05:18:13.47148 > admin.137: UDP, length 50
05:18:13.47148 > admin.44779 > ff02::1:3:5355: UDP, length 46
05:18:13.47148 > admin.44779 > 224.0.0.252:5355: UDP, length 46
05:18:13.47148 > admin.137: UDP, length 176
05:18:13.47148 > admin.44779: UDP, length 94
05:18:13.47148 > admin.137: ICMP admin udp port 44779 over
46 138
05:18:13.47148 P admin.47375 > ff02::1:3:5355: UDP, length 90
05:18:13.47148 P admin.47375 > 224.0.0.252:5355: UDP, length 90
05:18:13.47148 P > 192.168.168.255:137: UDP, length
05:18:13.47148 P admin.47375 > ff02::1:3:5355: UDP, length 90
05:18:13.47148 P admin.47375 > 224.0.0.252:5355: UDP, length 90
05:18:13.47148 P > 192.168.168.255:137: UDP, length
05:18:13.47148 P
```

<http://www.winpcap.org>

TCPdump tools

Tcpdump

```
tcpdump -c 20 -s 0 -i eth1 -A host 192.168.1.1 and tcp port http
```

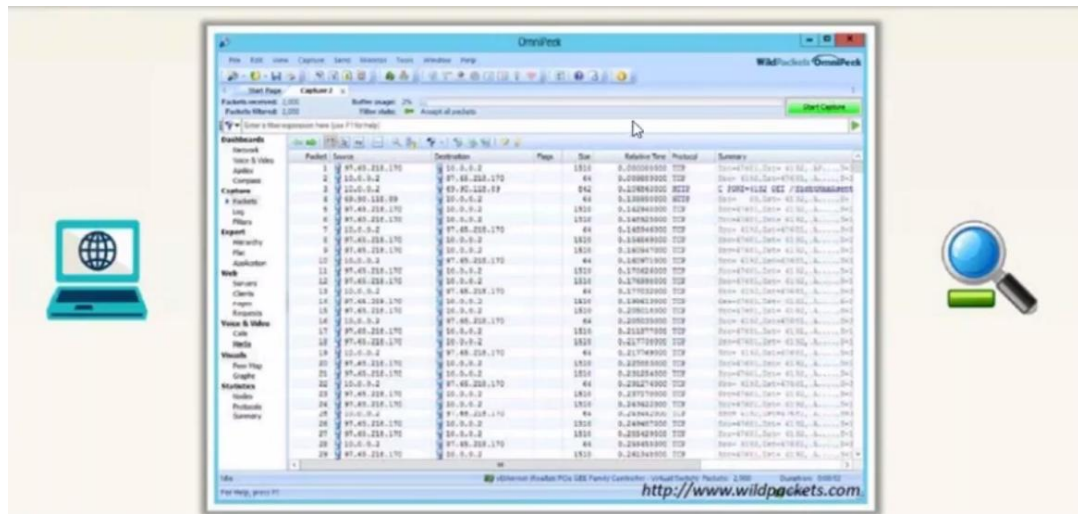
The parameter breakdown:

- c 20: Exit after capturing 20 packets.
- s 0: Don't limit the amount of payload data that is printed out. Print it all.
- i eth1: Capture packets on interface eth1
- A: Print packets in ASCII.
- host 192.168.1.1: Only capture packets coming to or from 192.168.1.1.
- and tcp port http: Only capture HTTP packets.

The figure below shows the different parts of the tcpdump output.



omniPeek network analyzer



- OmniPeek Network Analyzer is powerful tools for **network troubleshooting and analysis**.
- It offers features such as **packet capture and analysis, deep packet inspection, real-time monitoring, and visualization**.
- It supports multiple protocols, such as **Ethernet, IP, TCP, UDP, ICMP, and more**.
- It also has the ability to analyze **wireless networks** and provides detailed reports for analysis.

Scan open port on Linux for TCP protocol

```
(kali@kali)-[~]  
$ ss -tl
```

State	Recv-Q	Send-Q	Local Address:Port	Peer Address:Port	Process
LISTEN	0	1024	0.0.0.0:8834	0.0.0.0:*	
LISTEN	0	10	0.0.0.0:telnet	0.0.0.0:*	
LISTEN	0	244	127.0.0.1:postgresql	0.0.0.0:*	
LISTEN	0	244	127.0.0.1:5433	0.0.0.0:*	
LISTEN	0	244	127.0.0.1:5434	0.0.0.0:*	
LISTEN	0	1024	:::8834	:::*	
LISTEN	0	2	:::1:3350	:::*	
LISTEN	0	244	:::1:postgresql	:::*	
LISTEN	0	244	:::1:5433	:::*	
LISTEN	0	244	:::1:5434	:::*	
LISTEN	0	2	:::ms-wbt-server	:::*	

- -t tcp protocol
- L stands for listening sockets
- ss netstat command popular

Scan open port for UDP protocol

```
(kali@kali)-[~]  
$ ss -ul  
State      Recv-Q    Send-Q    Local Address:Port    Peer Address:Port    Process  
UNCONN     0          0          0.0.0.0:bootpc         0.0.0.0:*
```

- State: unconn equals unconnected
- We should monitor the actively listening

Use grep tool to search listen ports

```
(kali@kali)-[~]  
$ ss -tul | grep LISTEN  
tcp  LISTEN 0      1024      0.0.0.0:8834      0.0.0.0:*  
tcp  LISTEN 0       10      0.0.0.0:telnet    0.0.0.0:*  
tcp  LISTEN 0      244      127.0.0.1:postgresql 0.0.0.0:*  
tcp  LISTEN 0      244      127.0.0.1:5433    0.0.0.0:*  
tcp  LISTEN 0      244      127.0.0.1:5434    0.0.0.0:*  
tcp  LISTEN 0     1024      [::]:8834        [::]:*  
tcp  LISTEN 0        2      [::1]:3350        [::]:*  
tcp  LISTEN 0      244      [::1]:postgresql [::]:*  
tcp  LISTEN 0      244      [::1]:5433        [::]:*  
tcp  LISTEN 0      244      [::1]:5434        [::]:*  
tcp  LISTEN 0        2          *:ms-wbt-server  *:*
```

- Scan for listen port on TCP and UDP

Use grep tool to search listen ports

```
(kali㉿kali)-[~]  
$ ss -tuln | grep LISTEN  
tcp    LISTEN 0      1024      0.0.0.0:8834      0.0.0.0:*  
tcp    LISTEN 0       10      0.0.0.0:23       0.0.0.0:*  
tcp    LISTEN 0      244      127.0.0.1:5432    0.0.0.0:*  
tcp    LISTEN 0      244      127.0.0.1:5433    0.0.0.0:*  
tcp    LISTEN 0      244      127.0.0.1:5434    0.0.0.0:*  
tcp    LISTEN 0     1024      [::]:8834        [::]:*  
tcp    LISTEN 0        2      [::1]:3350        [::]:*  
tcp    LISTEN 0      244      [::1]:5432        [::]:*  
tcp    LISTEN 0      244      [::1]:5433        [::]:*  
tcp    LISTEN 0      244      [::1]:5434        [::]:*  
tcp    LISTEN 0        2          *:3389           *:*  
  
(kali㉿kali)-[~]  
$
```

- -n to **display only** the **port number** not resolve the name of the service

Display the process uses the open port number

```
(kali㉿kali)-[~]  
$ sudo ss -tulnp | grep LISTEN  
[sudo] password for kali:  
tcp    LISTEN 0      1024      0.0.0.0:8834      0.0.0.0:*      users:(("nessusd",pid=622,fd=18))  
tcp    LISTEN 0       10      0.0.0.0:23       0.0.0.0:*      users:(("inetutils-inetd",pid=604,fd=4))  
tcp    LISTEN 0      244      127.0.0.1:5432    0.0.0.0:*      users:(("postgres",pid=682,fd=4))  
tcp    LISTEN 0      244      127.0.0.1:5433    0.0.0.0:*      users:(("postgres",pid=704,fd=6))  
tcp    LISTEN 0      244      127.0.0.1:5434    0.0.0.0:*      users:(("postgres",pid=669,fd=6))  
tcp    LISTEN 0     1024      [::]:8834        [::]:*      users:(("nessusd",pid=622,fd=22))  
tcp    LISTEN 0        2      [::1]:3350        [::]:*      users:(("xrdp-sesman",pid=630,fd=7))  
tcp    LISTEN 0      244      [::1]:5432        [::]:*      users:(("postgres",pid=682,fd=3))  
tcp    LISTEN 0      244      [::1]:5433        [::]:*      users:(("postgres",pid=704,fd=5))  
tcp    LISTEN 0      244      [::1]:5434        [::]:*      users:(("postgres",pid=669,fd=5))  
tcp    LISTEN 0        2          *:3389          *:~      users:(("xrdp",pid=655,fd=11))
```

- 'users', i.e., the processes which use the port number

Ssh commands

```
kali@kali:~$ sudo service ssh
Usage: /etc/init.d/ssh {start|stop|reload|force-reload|restart|try-restart|status}.
kali@kali:~$
```

```
kali@kali:~$ sudo service ssh
Usage: /etc/init.d/ssh {start|stop|reload|force-reload|restart|try-restart|status}.
kali@kali:~$ sudo service ssh status
● ssh.service - OpenBSD Secure Shell server
   Loaded: loaded (/lib/systemd/system/ssh.service; disabled; vendor preset: disabled)
   Active: inactive (dead)
     Docs: man:sshd(8)
           man:sshd_config(5)
kali@kali:~$
```

Enable the ssh service

```
kali@kali:~$ sudo service enable ssh
enable: unrecognized service
kali@kali:~$ sudo systemctl enable ssh
Synchronizing state of ssh.service with SysV service script with /lib/systemd/systemd-sysv-in
stall.
Executing: /lib/systemd/systemd-sysv-install enable ssh
Created symlink /etc/systemd/system/ssh.service → /lib/systemd/system/ssh.service.
Created symlink /etc/systemd/system/multi-user.target.wants/ssh.service → /lib/systemd/system
/ssh.service.
kali@kali:~$ sudo service ssh start
kali@kali:~$ sudo service ssh status
● ssh.service - OpenBSD Secure Shell server
   Loaded: loaded (/lib/systemd/system/ssh.service; enabled; vendor preset: disabled)
   Active: active (running) since Tue 2022-01-04 12:51:06 EST; 4s ago
     Docs: man:sshd(8)
           man:sshd_config(5)
  Process: 6407 ExecStartPre=/usr/sbin/sshd -t (code=exited, status=0/SUCCESS)
 Main PID: 6408 (sshd)
    Tasks: 1 (limit: 4639)
   Memory: 1.2M
    CGroup: /system.slice/ssh.service
           └─6408 sshd: /usr/sbin/sshd -D [listener] 0 of 10-100 startups

Jan 04 12:51:06 kali systemd[1]: Starting OpenBSD Secure Shell server ...
Jan 04 12:51:06 kali sshd[6408]: Server listening on 0.0.0.0 port 22.
Jan 04 12:51:06 kali sshd[6408]: Server listening on :: port 22.
Jan 04 12:51:06 kali systemd[1]: Started OpenBSD Secure Shell server.
kali@kali:~$
```

From Ubuntu I login to kali remotely

```
ubuntu@ubuntu:~$ ssh kali@192.168.30.135
The authenticity of host '192.168.30.135 (192.168.30.135)' can't be established
.
ECDSA key fingerprint is SHA256:gvAtIL03jQho3VXm7ck0tntbQyaXm0M1K0xY/D5xTXw.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '192.168.30.135' (ECDSA) to the list of known hosts.
kali@192.168.30.135's password:
kali@kali:~$ whoami
kali
kali@kali:~$ sudo whoami
[sudo] password for kali:
root
kali@kali:~$
```

Search about application want to download

```
kali@kali:~$ apt-cache search apache2
apache2 - Apache HTTP Server
apache2-bin - Apache HTTP Server (modules and other binary files)
apache2-data - Apache HTTP Server (common files)
apache2-dev - Apache HTTP Server (development headers)
apache2-doc - Apache HTTP Server (on-site documentation)
apache2-ssl-dev - Apache HTTP Server (mod_ssl development headers)
apache2-suexec-custom - Apache HTTP Server configurable suexec program for mod_suexec
apache2-suexec-pristine - Apache HTTP Server standard suexec program for mod_suexec
apache2-utils - Apache HTTP Server (utility programs for web servers)
cortado - streaming applet for Ogg formats
dehydrated-apache2 - dehydrated challenge response support for Apache2
dwww - Read all on-line documentation with a WWW browser
gitweb - fast, scalable, distributed revision control system (web interface)
libapache-mod-jk-doc - Documentation of libapache2-mod-jk package
libapache2-authcasimple-perl - Apache2 module to authenticate through a CAS server
libapache2-authcookie-perl - Perl Authentication and Authorization via cookies
libapache2-mod-apparmor - changehat AppArmor library as an Apache module
libapache2-mod-apreq2 - generic Apache request library - Apache module
libapache2-mod-authn-gss - GSS authentication module for Apache2
```

Give me information about this application

```
kali@kali:~$ apt show apache2
Package: apache2
Version: 2.4.51-2
Priority: optional
Section: httpd
Maintainer: Debian Apache Maintainers <debian-apache@lists.debian.org>
Installed-Size: 633 kB
Provides: httpd, httpd-cgi
```


Install information

```
kali@kali:~$ sudo apt install apache2
Reading package lists... Done
Building dependency tree
Reading state information... Done
The following packages were automatically installed and are no longer required:
  doc-base galera-3 gir1.2-packagekitglib-1.0 greenbone-security-assistant
  greenbone-security-assistant-common gvm-tools gvmd gvmd-common libappstream4
```

Remove application and stay the configuration files

```
kali@kali:~$  
kali@kali:~$ sudo apt remove apache2
```

Remove the package ssh and configuration file

```
kali@kali:~$  
kali@kali:~$ sudo apt remove --purge apache2
```

Application does not found with apt

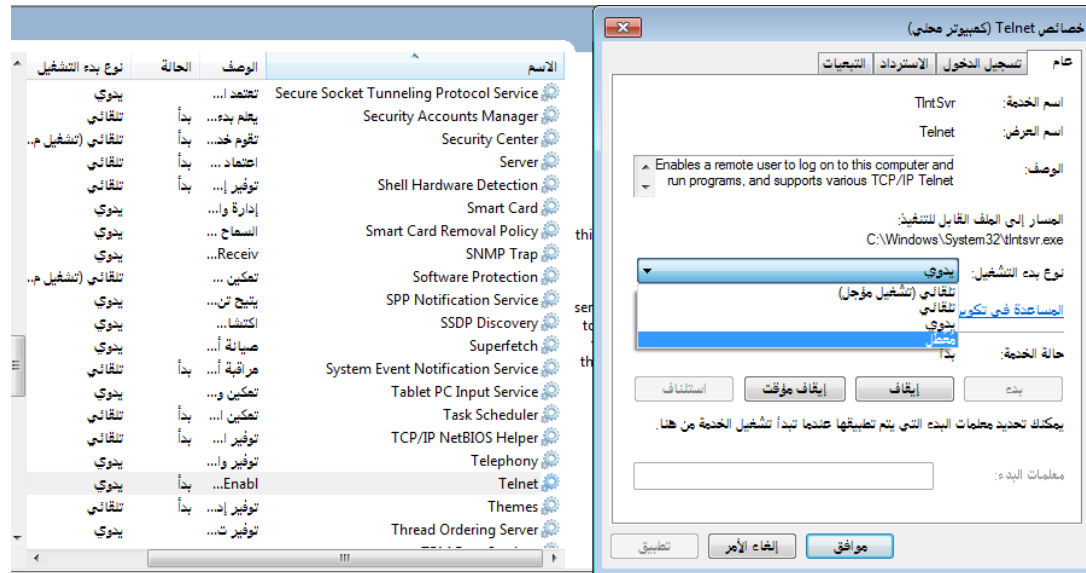
- `Dpkg -l [name of the application]`

telnet to window not replay

```
-p port          local port number
(kali~ kali)-[~]
$ telnet 192.168.30.138
Trying 192.168.30.138 ...
telnet: Unable to connect to remote host: Connection refused

-t             answer TELNET negotiation
(kali~ kali)-[~]
$             UDP mode
-v            verbose [use twice to be more v
-w secs       timeout for connects and final
```

Enable telnet on window 7



Open port

```
(kali㉿kali)-[~]  
$ telnet 192.168.30.138  
Trying 192.168.30.138 ...  
Connected to 192.168.30.138.  
Escape character is '^]'.  
ls  
Welcome to Microsoft Telnet Service  
login: PC  
password:r  
??  
Login Failed  
login: PC  
password:v
```

allow broadcasts
source-routing hop point[s], up to 9
source-routing pointer: 4, 8, 16, 20
this cruft
delay interval for lines sent, in seconds
set keepalive option on socket if non-zero
listen mode, for inbound connections
numeric-only IP addresses, no DNS lookups
hex dump of traffic if non-zero
local port number
randomize local and remote port numbers
quit after EOF on stdin and delete buffers
local source address
set Type Of Service for outgoing connections
answer TELNET negotiation
UDP mode
verbose [use twice to be more verbose]

Vulnerability

- **Vulnerability Assessment:** Scans and tests for vulnerabilities but does not intentionally exploit them

Vulnerability Categories

- **Misconfiguration**: improperly configuring a service or application
- **Default installation**: failure to **change settings** in an **application** that come by default
- **Buffer overflow**: code execution flaw
- **Missing patches**: systems that **have not been patched**
- **Design flaws**: **flaws** inherent to **system design** such as **encryption and data validation**
- **Operating System Flaws**: flaws specific to each OS
- **Default passwords**: **leaving default passwords** that come with system/application

Vulnerability Scanning tools

- Industry standard is **Tenable's Nessus**.
- **GFI LanGuard**.
- **Nikto** - CLI; is a **web server assessment tool**. It is designed to find various **default and insecure files, configurations and programs** on any type of web server.
- **OpenVAS** - Best competitor to Nessus and is free.
- **wpscan** - CLI; Scan **WordPress** websites.
- **MBSA** - **Microsoft Baseline Security Analyzer**.
- **FreeScan** - Well known for **testing websites and applications**.
- **Qualys** is a framework

Common Vulnerability Scanning Tools

- Industry standard is **Tenable's Nessus**.
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CVE – Common Vulnerabilities and Exposures



Common Vulnerabilities and Exposures

The Standard for Information Security Vulnerability Names

[Home](#) | [CVE IDs](#) | [About CVE](#) | [Compatible Products & More](#) | [Community News](#) | [Site Search](#)

TOTAL CVE IDs: **78792**

[HOME](#) > [CVE LIST](#)

Section Menu

CVE IDs

[Coverage Goals](#)
[Reference Key/Maps](#)
[Updates & Feeds](#)

CVE List (all existing CVE IDs)

[Downloads](#)
[Search CVE List](#)
[Search tips](#)
[View Entire CVE List \(html\)](#)
[NVD Advanced CVE Search](#)
[CVE ID Scoring Calculator](#)

Request a CVE ID

[CVE Numbering Authorities](#)

CVE IDs

The [CVE List Master Copy](#) is hosted on this CVE website. The [U.S. National Vulnerability Database \(NVD\)](#), which is built upon and fed by the CVE List, provides enhanced information about CVE IDs. Learn more about the [CVE and NVD relationship](#).

What would you like to do?

Data Feeds

[Available via
Purdue
University & NVD](#)

Request a

**CVE ID
number**

[Click for
guidelines &
more](#)

- Is a list of publicly disclosed vulnerabilities and exposures that is ***maintained by MITRE***.
- Such as vulnerable called **Log4Shell** take the code CVE-2021-44228
- Header CVE
- 2021 year discovered
- 44228 unique id

CVSS - Common Vulnerability Scoring System

Places numerical score based on severity

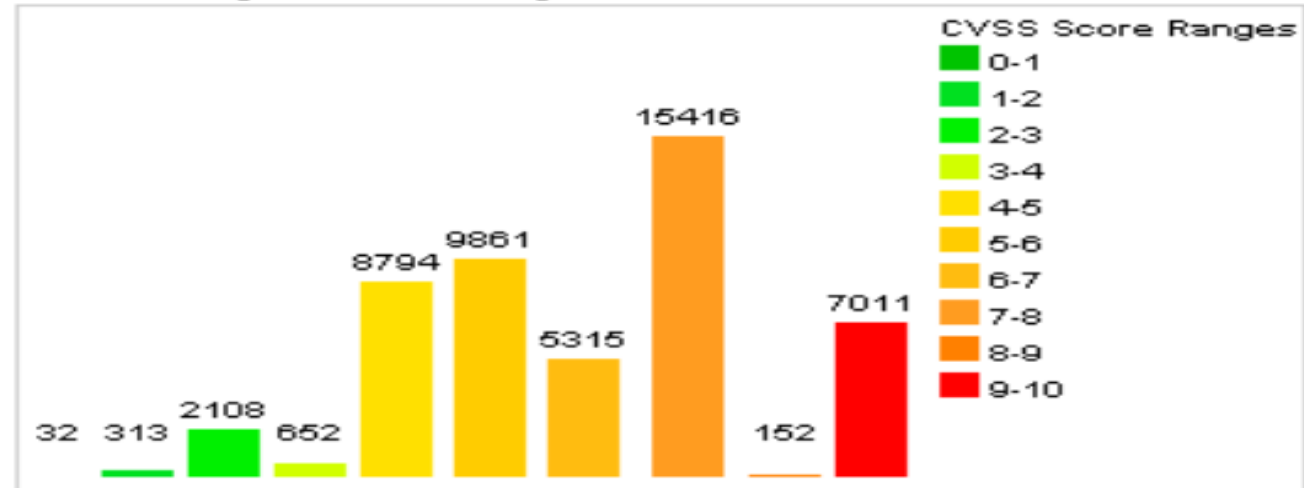
Distribution of all vulnerabilities by CVSS Scores

CVSS Score	Number Of Vulnerabilities	Percentage
0-1	32	0.10
1-2	313	0.60
2-3	2108	4.20
3-4	652	1.30
4-5	8794	17.70
5-6	9861	19.90
6-7	5315	10.70
7-8	15416	31.00
8-9	152	0.30
9-10	7011	14.10
Total	49654	

Weighted Average CVSS Score: **6.9**

- None - white (0.0)
- Low - green tones (0.1 - 3.9)
- Medium - yellow/light orange (4.0 - 4.9)
- High - orange (7.0 - 8.0)
- Critical - red (9.0 - 10.0)

Vulnerability Distribution By CVSS Scores



NVD - National Vulnerability Database

- **National Vulnerability Database (NVD)**
- NVD is a **database**, maintained by *NIST*
- **Fully synchronized** with the **MITRE CVE** list;
- **US Gov.** vulnerabilities **repository**.
- This **utilizes CVE identifiers** and supplies **additional information** on vulnerabilities **such as CVSS threat scores, technical details, affected entities**, and **resources** for further investigation.

Ping Sweep Countermeasures

- Here are ***six ways*** to protect your network from ping sweep attacks:
- **Configure firewalls** to block incoming ICMP echo requests from unknown or untrusted sources.
- **Use intrusion detection systems (IDSes) and intrusion prevention systems (IPSeS)**, such as Snort, to detect and prevent ping sweep attempts.
- **Carefully evaluate the type of ICMP traffic** flowing through enterprise networks.
- **Terminate the connection** with any host sending more than 10 ICMP ECHO requests.
- **Use a DMZ** and allow only commands such as ICMP ECHO_REPLY, HOST UNREACHABLE, and TIME EXCEEDED in the DMZ.
- **Limit ICMP traffic** with access-control lists (ACLs) to the ISP's specific IP addresses.

IP Spoofing Countermeasures

- Here are several ways to defend against IP spoofing attacks:
- **Encrypt all network traffic** using cryptographic network protocols such as IPsec, TLS, SSH, and HTTPS.
- **Use multiple firewalls** to provide a multi-layered depth of protection.
- **Do not rely on IP-based authentication.**
- **Use a random initial sequence number** to prevent IP spoofing attacks based on sequence number spoofing.
- **Ingress Filtering:** Use routers and firewalls at the network's perimeter to filter incoming packets that appear to be from an internal IP address.
- **Egress Filtering:** Filter all outgoing packets that have an invalid local IP address as the source address.

Port Scanning Countermeasures

- Here are **eight ways to protect your network** from port scanning:
- **Configure firewall and IDS rules** to detect and block probes.
- **Run port scanning tools** against your network to check if your firewall properly detects port scanning activity.
- **Ensure that the mechanisms** used for routing and filtering at the routers and firewalls cannot be bypassed using specific source ports or routing methods.
- **Update the firmware** for your **routers, IDS, and firewalls** to their latest versions.
- **Use a custom rule set** on your firewall to **lock down** the network and **block unwanted** ports.
- **Filter all ICMP messages** (both inbound and outbound) at the firewalls and routers.
- **Perform TCP and UDP scanning** along with ICMP probes against your organization's IP address space to check your network configuration and available ports.
- **Ensure that anti-scanning and anti-spoofing rules** are properly configured.

Banner Grabbing Countermeasures

- **Disabling or Changing Banner**

- You can display **false banners** to **mislead** or deceive attackers.
- **Turn off any unnecessary services** on the network host to limit the disclosure of information.
- Use **server masking** tools to **disable** or **change banner information**.
- For Apache 2.x with the *mod_headers* module, use a directive in the *httpd.conf* file to **change the banner information header** and set the server as a **new server name**.
- Alternatively, you can change the *ServerSignature* line to *ServerSignature* Off in the *httpd.conf* file.

Banner Grabbing Countermeasures

- **Hiding File Extensions from Web Pages**
 - **File extensions** *reveal* information about the **underlying server technology** that an attacker can use to launch attacks.
 - **Hide file** extensions to **mask web technologies**.
 - You can replace **application mappings** such as *.asp* with *.htm* or *.foo*, to **disguise the identities** of servers.
 - Apache users can use **mod_negotiation** directives.
 - **It is preferable to not use file extensions at all.**

Modern Cybersecurity Reconnaissance & Analysis Tools

- Part A — Network Enumeration
 - RustScan , enum4linux-ng , DNSRecon
- Part B — Web Enumeration
 - ffuf , Burp Suite , Nikto
- Part C — Network Analysis
 - NetworkMiner , TCPdump
- Part D — Tool Comparison
- Compare between:
 - Nmap vs RustScan
 - WhatWeb vs cURL
 - Wireshark vs NetworkMiner
 - Nmap vs enum4linux-ng

AI-assisted Analysis

- Take the result “Nmap -A [IP]” use : **ChatGPT / Gemini / Claude to find** AI Finding → Evidence → My Verification
- AI + Wireshark use .pcap file and use the AI to analysis the events in the file